

Dynamic Mode Decompositions and Vector Autoregressions *

Thomas J. Sargent, Yatheesan J. Selvakumar, and Ziyue Yang

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Abstract

We establish connections between dynamic mode decompositions (DMDs), vector autoregressions, and linear state-space models, showing that DMD provides a computationally efficient, SVD-based estimator of low-rank first-order VAR projection coefficients in high-dimensional settings. When the measurement matrix has full column rank, the recovered nonzero eigenvalues coincide with those of the underlying state transition matrix. We apply DMD to a 100-household heterogeneous-agent economy with complete markets and Gorman aggregation. From high-dimensional household income and consumption panels, DMD successfully recovers low-dimensional aggregate dynamics: estimated modes track latent aggregate states with correlations exceeding 0.90, and cross-sectional loadings reveal the sharing rule governing redistribution. This demonstrates DMD's capacity to extract economically meaningful low-dimensional structure from microeconomic panels.

Keywords: Vector Autoregression, LQG State-Space Model, Innovations Representation, Singular Value Decomposition, Dynamic Mode Decomposition, Information Projection

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1 Introduction

Applied mathematicians have recently used Dynamic Mode Decompositions (DMDs) as a “machine learning” technique to quantify non-random dynamical systems, including fluid dynamics (e.g., see [Brunton and Kutz \(2022\)](#)). This paper instead uses DMDs to quantify a dynamical system that is persistently buffeted by random shocks. We describe connections among dynamic mode decompositions (DMDs), vector autoregressions (VARs), and a special case of what [Stock and Watson \(2016\)](#) call dynamic factor models. DMDs are calculated using singular value decompositions and associated eigendecompositions, but of different objects than the ones that underlie the dynamic factor models described by [Stock and Watson](#). We connect a reduced-rank first-order VAR to an observation-noise-free linear state-space model. We show that the population first-order VAR projection coefficient matrix has the same reduced rank and can be efficiently estimated and represented by DMD. We use a Dynamic Mode Decomposition (DMD) to infer a reduced-rank first-order VAR from a data set $[\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_T, \mathbf{y}_{T+1}]$.

Through its connection to the [Koopman \(1931\)](#) operator, the DMD algorithm has also been used to approximate nonlinear dynamics (see [Williams et al. \(2015\)](#), [Brunton et al. \(2016\)](#), [Mezić \(2022\)](#)). Our model superficially resembles ones of [Geweke \(1977\)](#), [Sargent and Sims \(1977\)](#), [Geweke and Singleton \(1981\)](#), [Stock and Watson \(2002\)](#), [Bai \(2003\)](#), and [Bai and Ng \(2013\)](#), but our statistical model of hidden factors differs from theirs. Like those authors, we estimate “non-structural Kepler-stage” descriptive models that can compress data and reveal patterns. As [Koopmans \(1947\)](#) recommends, we too want eventually to interpret these empirical regularities with a “structural Newton-stage” model that is cast in terms of parameters that describe market structures and decision makers’ preferences, constraints, and information flows. Koopmans interpreted [Burns and Mitchell \(1946\)](#) as such a “Kepler-stage” model of business

cycles, in contrast to the structural, simultaneous stochastic difference equation models of business cycles that the tools developed by [Koopmans \(1950\)](#), [Hood and Koopmans \(1953\)](#), and [Marschak \(1953\)](#).

This paper is organized as follows. Section 2 describes the state-space representations of our statistical model, its relationship to a reduced-rank first-order VAR, and its connections to dynamic mode decompositions. This section focuses on five objects:

1. A statistical model for a covariance-stationary stochastic process $\{\mathbf{y}_t\}_{t=-\infty}^{\infty}$ that takes the form of a special observation-noise-free LQG state-space model, where $\mathbf{y}_t$ is an $M \times 1$ vector.
2. An innovations representation of that state-space model.
3. An infinite-order vector autoregression (VAR) for $\{\mathbf{y}_t\}_{t=-\infty}^{\infty}$.
4. A reduced-rank first-order VAR.
5. A dynamic mode decomposition (DMD) of a data set $[\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_T, \mathbf{y}_{T+1}]$.

Section 2 describes situations in which

- items [1] and [2] are both valid representations of item [3]
- items [3] and [4] coincide
- item [5] provides a good way to estimate item [4]

Having acknowledged that we understand that a DMD provides a descriptive statistical model, not what theoretical IO or macroeconomists call a “structural” model, Section 3 proceeds to analyze properties of our Section 2 model as a statistical approximation to a version of a structural dynamic stochastic general equilibrium (DSGE) model described by [Hansen and Sargent \(2013, ch. 12\)](#). The structural DSGE model describes competitive equilibrium prices and quantities for a heterogeneous-agent economy containing what Milton Friedman called “homogenizing mechanisms” that put a factor structure into the dynamics of cross-sections of households’ income and consumption rates. The equilibrium of our DSGE model has a representation that takes

the form of a linear state-space model and an associated likelihood function that via “information projection” techniques can be compared to its DMD counterpart for the same observable process.¹ The information-projection experiments in our Section 3 laboratory are designed to shed light on an empirical application to US CEX data by Sargent et al. (2025). Section 4 offers concluding remarks about promising applications. Appendix A describes how DMD differs from principal components analysis in dynamic factor models. In a nutshell, PCA constructs a singular value decomposition of a covariance matrix and from it constructs a “static” statistical representation, while DMD provides a spectral representation of the autoregressive coefficient matrix of a first-order reduced-rank vector autoregression. Appendices B, C, D, and E provide assorted technical details.

2 Statistical Model that Underlies DMD

Let $\mathbf{y}_t$ be an $M \times 1$ vector and let $\mathbf{y}^{t-1} := \{\mathbf{y}_{t-s}\}_{s=1}^{\infty}$. Throughout we use $\perp$ to denote orthogonality and work with centered observables with $\mathbb{E}[\mathbf{y}_t] = 0$. We take as our baseline statistical model an observation-noise-free linear state-space model with potentially rank-deficient measurement matrix $\mathbf{G}$:

$$\begin{aligned}\mathbf{x}_{t+1} &= \underset{N_x \times N_x}{\mathbf{A}} \mathbf{x}_t + \underset{N_x \times N_x}{\mathbf{C}} \mathbf{w}_{t+1} \\ \mathbf{y}_t &= \underset{M \times N_x}{\mathbf{G}} \mathbf{x}_t, \quad t \geq 0, \quad \mathbf{x}_0 \sim \mathcal{N}(0, \boldsymbol{\Omega}_{\mathbf{x}})\end{aligned}\tag{1}$$

where random shocks $\mathbf{w}_{t+1} \sim \mathcal{N}(0, \mathbf{I}_{N_x})$, and $\boldsymbol{\Omega}_{\mathbf{x}}$ is a positive semi-definite solution of the discrete Lyapunov equation

$$\boldsymbol{\Omega}_{\mathbf{x}} = \mathbf{A} \boldsymbol{\Omega}_{\mathbf{x}} \mathbf{A}^{\top} + \mathbf{C} \mathbf{C}^{\top} .\tag{2}$$

¹See footnote 18 below for an explanation of information projection. Fernández-Villaverde and Rubio-Ramírez (2004) provide an application.

Let $r := \text{rank}(\mathbf{G}) \leq \min\{M, N_x\}$. When $r < M$, $\{\mathbf{y}_t\}$ is supported on the r -dimensional subspace $\text{range}(\mathbf{G}) \subset \mathbb{R}^M$ and $\text{Cov}(\mathbf{y}_t)$ is singular. We assume $\mathbf{A}$ is stable so that $\{\mathbf{x}_t\}$ and $\{\mathbf{y}_t\}$ are covariance-stationary. The optimal one-step-ahead prediction conditioned on $\mathbf{x}_t$ is

$$\tilde{\mathbf{x}}_{t+1} = \mathbb{E}[\mathbf{x}_{t+1} \mid \mathbf{x}_t] = \mathbf{A} \mathbf{x}_t,$$

and the associated one-step-ahead prediction error covariance matrix is

$$\mathbb{E} [(\mathbf{x}_{t+1} - \tilde{\mathbf{x}}_{t+1})(\mathbf{x}_{t+1} - \tilde{\mathbf{x}}_{t+1})^\top] = \mathbf{C} \mathbf{C}^\top.$$

The associated steady-state innovations representation is cast in terms of the one-step-ahead prediction vector²

$$\hat{\mathbf{x}}_{t+1} = \mathbb{E}[\mathbf{x}_{t+1} \mid \mathbf{y}^t].$$

This innovations representation is

$$\begin{aligned} \hat{\mathbf{x}}_{t+1} &= \underset{N_x \times 1}{\mathbf{A}} \underset{N_x \times N_x}{\hat{\mathbf{x}}_t} + \underset{N_x \times M}{\mathbf{K}} \underset{M \times 1}{\mathbf{a}_t}, \\ \mathbf{y}_t &= \underset{M \times 1}{\mathbf{G}} \underset{M \times N_x}{\hat{\mathbf{x}}_t} + \mathbf{a}_t, \end{aligned} \tag{3}$$

where $\mathbf{a}_t = \mathbf{y}_t - \mathbb{E}[\mathbf{y}_t \mid \mathbf{y}^{t-1}]$, $\mathbf{a}_t \perp \mathbf{y}^{t-1}$, and the steady-state Kalman gain is

$$\mathbf{K} = \mathbf{A} \boldsymbol{\Sigma}_\infty \mathbf{G}^\top (\mathbf{G} \boldsymbol{\Sigma}_\infty \mathbf{G}^\top)^+, \tag{4}$$

where $(\cdot)^+$ denotes the Moore–Penrose pseudoinverse³, and

$$\boldsymbol{\Sigma}_\infty = \mathbb{E} [(\mathbf{x}_t - \hat{\mathbf{x}}_t)(\mathbf{x}_t - \hat{\mathbf{x}}_t)^\top]$$

²For connections between a steady-state innovations representation and a state-space representation like (1), see Hansen and Sargent (2013, ch. 9), Ljungqvist and Sargent (2018, ch. 2), and Fernández-Villaverde et al. (2007).

³We use a pseudoinverse in (4) because $\mathbf{G} \boldsymbol{\Sigma}_\infty \mathbf{G}^\top$ has rank at most $r := \text{rank}(\mathbf{G})$ and is singular when $r < M$.

is the positive semidefinite solution of the following algebraic matrix Riccati equation:

$$\Sigma_\infty = \mathbf{C} \mathbf{C}^\top + (\mathbf{A} - \mathbf{K} \mathbf{G}) \Sigma_\infty (\mathbf{A} - \mathbf{K} \mathbf{G})^\top. \quad (5)$$

Consider the population linear least-squares projection of $\mathbf{y}_t$ on $\mathbf{y}_{t-1}$

$$\begin{aligned} \mathbf{y}_t &= \mathbf{B}_{M \times M}^* \mathbf{y}_{t-1} + \boldsymbol{\varepsilon}_t, \\ \mathbb{E}[\boldsymbol{\varepsilon}_t \mathbf{y}_{t-1}^\top] &= \mathbf{0}. \end{aligned} \quad (6)$$

where least-squares orthogonality conditions imply that

$$\mathbf{B}^* = \text{Cov}(\mathbf{y}_t, \mathbf{y}_{t-1}) \text{Cov}(\mathbf{y}_{t-1})^+. \quad (7)$$

Note that when $\text{Cov}(\mathbf{y}_{t-1})$ is singular, there are many matrices that yield the same fitted values on $\text{range}(\text{Cov}(\mathbf{y}_{t-1}))$. We adopt the unique Moore–Penrose representative. When the restriction $\mathbb{E}[\boldsymbol{\varepsilon}_t \mathbf{y}_{t-j}^\top] = \mathbf{0}$ holds for all $j \geq 1$, the first-order VAR (6) becomes an infinite-order VAR

$$\mathbf{y}_t = \sum_{j=1}^{\infty} \mathbf{B}_j \mathbf{y}_{t-j} + \boldsymbol{\varepsilon}_t,$$

with $\mathbf{B}_1 = \mathbf{B}^*$ and $\mathbf{B}_j = \mathbf{0} \ \forall j \geq 2$.

Let $\boldsymbol{\Omega}_x := \text{Cov}(\mathbf{x}_t)$ denote the stationary state covariance that solves the Lyapunov equation (2). In general, the least-squares projection coefficient matrix satisfies⁴

$$\mathbf{B}^* = \mathbf{G} \mathbf{A} \Sigma_x \mathbf{G}^\top (\mathbf{G} \Sigma_x \mathbf{G}^\top)^+, \quad \mathbf{B}^* \mathbf{v} = \lambda \mathbf{v}, \lambda \neq 0 \Rightarrow \mathbf{v} \in \text{range}(\mathbf{G}). \quad (8)$$

For the rest of this section, we focus on the case where the measurement map is injective, i.e., the measurement matrix $\mathbf{G}$ has full column rank ($\text{rank}(\mathbf{G}) = N_x$, so

⁴See Appendix C.1.

$N_x \leq M$).⁵ The following proposition characterizes the induced VAR(1) and the link between the population projection coefficient matrix $\mathbf{B}^*$ and the state transition matrix $\mathbf{A}$ in representation (1).

Proposition 1. *Suppose that $\mathbf{G}$ has full column rank and let $\mathbf{G}^+ = (\mathbf{G}^\top \mathbf{G})^{-1} \mathbf{G}^\top$. Define*

$$\bar{\mathbf{B}} := \mathbf{G} \mathbf{A} \mathbf{G}^+, \quad \mathbf{u}_t := \mathbf{G} \mathbf{C} \mathbf{w}_t.$$

Then the following statements hold:

(i) $\{\mathbf{y}_t\}$ has representation

$$\mathbf{y}_t = \bar{\mathbf{B}} \mathbf{y}_{t-1} + \mathbf{u}_t,$$

where $\{\mathbf{u}_t\}$ is i.i.d. with $\mathbb{E}[\mathbf{u}_t] = \mathbf{0}$ and $\mathbf{u}_t$ is orthogonal to $\mathbf{y}^{t-1}$. Moreover, $\mathbf{a}_t$ in (3) satisfies $\mathbf{a}_t = \mathbf{u}_t$, and $\mathbf{B}^$ in (7) satisfies*

$$\mathbf{B}^* \mathbf{y}_{t-1} = \bar{\mathbf{B}} \mathbf{y}_{t-1} \quad a.s.$$

(ii) *If in addition $\boldsymbol{\Omega}_{\mathbf{x}} := \text{Cov}(\mathbf{x}_t) \succ 0$, then*

$$\mathbf{B}^* = \bar{\mathbf{B}} = \mathbf{G} \mathbf{A} \mathbf{G}^+, \quad \mathbf{A} = \mathbf{G}^+ \mathbf{B}^* \mathbf{G}.$$

Proof. See Appendix C.5.1. □

A consequence is that when $\mathbf{G}$ has full column rank, the nonzero eigenvalues of $\mathbf{B}^*$ equal the eigenvalues of $\mathbf{A}$. When $M > N_x$, $\mathbf{B}^*$ also has some zero eigenvalues. Furthermore, when $\mathbf{C} \mathbf{C}^\top \succ 0$, the following identities link the innovations representation

⁵We leave treatment of the rank-deficient case to Appendix C.1.

with the state-space representation and $\mathbf{B}^*$.⁶

$$\Sigma_\infty = \mathbf{C} \mathbf{C}^\top, \quad \mathbf{A} = \mathbf{K} \mathbf{G}, \quad \mathbf{B}^* = \mathbf{G} \mathbf{K}. \quad (9)$$

Having characterized the population projection coefficient matrix $\mathbf{B}^*$, we now proceed to study the properties of DMD as a reduced-rank statistical estimator of $\mathbf{B}^*$. We will proceed according to the following road map. Recall that our state-space model (1) assumes an observation-noise-free measurement equation $\mathbf{y}_t = \mathbf{G} \mathbf{x}_t$ and a stable transition matrix $\mathbf{A}$. We mainly focus on the case in which the measurement matrix $\mathbf{G}$ has full column rank ($\text{rank}(\mathbf{G}) = N_x$). Under this condition, Proposition 1(i) establishes that the first-order and infinite-order VAR representations of $\{\mathbf{y}_t\}$ coincide and that the least-squares projection coefficient matrix $\bar{\mathbf{B}} = \mathbf{G} \mathbf{A} \mathbf{G}^+$. Adding the assumption $\Sigma_x \succ 0$ in Proposition 1(ii) ensures uniqueness of the projection coefficient $\mathbf{B}^* = \bar{\mathbf{B}}$.

In Section 2.2, we assume that $N = N_x$, so that the number of the assumed DMD modes matches the dimension of $\mathbf{x}_t$. The conditions that $\text{rank}(\mathbf{G}) = N_x$ and that Φ has full column rank ensure that DMD representation (17) below aligns with our innovations representation (3).

In our information-projection laboratory in Section 3, even though $N = N_x$ holds, the full-column-rank condition $\text{rank}(\mathbf{G}) = N_x$ can be violated. However, as we will show in Section 3.5, DMD can still extract underlying hidden states and modes. Appendix D constructs a reduced-rank synthetic hidden state such that $\mathbf{G}$ has full column rank in the reduced synthetic state space, and derives additional properties when the full-column-rank condition holds.

⁶For detailed proofs, see Appendix C.2

2.1 DMD as Estimator of Reduced-Rank First-Order VAR

In this section, we assume that $\{\mathbf{y}_t\}_{t=0}^\infty$ is generated by model (1). We show how a dynamic mode decomposition (DMD) procedure provides a computationally efficient rank- N estimator $\hat{\mathbf{B}}$ of $\mathbf{B}^*$.

The data set is organized as follows. Let $\mathbf{y}_t$ denote an $M \times 1$ vector of demeaned random variables for $t = 1, \dots, T+1$,⁷ and assume $M > T$, so there are more variables each period than time periods in the data set. Stack the $\mathbf{y}_t$'s across time to create two $M \times T$ data matrices $\mathbf{Y}$ and $\mathbf{Y}'$:

$$\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_T], \quad (10)$$

$$\mathbf{Y}' = [\mathbf{y}_2, \mathbf{y}_3, \dots, \mathbf{y}_{T+1}]. \quad (11)$$

When $M > T$, the least-squares estimator of the projection of $\mathbf{y}_t$ on $\mathbf{y}_{t-1}$ is underdetermined. DMD resolves this by first compressing the matrix $\mathbf{Y}$ to an N -dimensional subspace and then applying least squares on that reduced space. Fix an assumed rank $N \leq \min\{M, T\}$. Compute the rank- N truncated SVD of $\mathbf{Y}$ with $\mathbf{U} \in \mathbb{R}^{M \times N}$, $\mathbf{\Sigma} = \text{diag}(\sigma_1, \dots, \sigma_N) \in \mathbb{R}^{N \times N}$, and $\mathbf{V} \in \mathbb{R}^{T \times N}$:

$$\mathbf{Y} \approx \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top, \quad (12)$$

which is the best rank- N approximation of $\mathbf{Y}$ in the Frobenius norm.⁸ Furthermore, $\mathbf{U}^\top \mathbf{U} = \mathbf{I}_N$, $\mathbf{V}^\top \mathbf{V} = \mathbf{I}_N$, and $\mathbf{U}^\top \mathbf{Y} = \mathbf{\Sigma} \mathbf{V}^\top$. Define the rank- N approximation $\mathbf{Y}_N := \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top$. DMD seeks an estimator $\hat{\mathbf{B}}$ of the first-order VAR coefficient matrix that satisfies

$$\hat{\mathbf{B}} \in \arg \min_{\text{rank}(\mathbf{B}) \leq N} \|\mathbf{Y}' - \mathbf{B} \mathbf{Y}_N\|_F^2, \quad (13)$$

⁷Hirsh et al. (2020) show that centering the data is equivalent to incorporating an affine term in the dynamic model and improves the performance of DMD in correctly extracting the dynamics of the data.

⁸This follows from the Eckart–Young–Mirsky theorem; see (Golub and Van Loan, 2013, Theorem 2.4.8).

where $\|\cdot\|_F$ denotes the Frobenius norm. Since $\mathbf{Y}_N$ has rank N , its Moore–Penrose pseudoinverse is $\mathbf{Y}_N^+ = \mathbf{V} \mathbf{\Sigma}^{-1} \mathbf{U}^\top$. DMD selects the following truncated-SVD solution to (13):

$$\hat{\mathbf{B}} := \mathbf{Y}' \mathbf{Y}_N^+ = \mathbf{Y}' \mathbf{V} \mathbf{\Sigma}^{-1} \mathbf{U}^\top, \quad (14)$$

which has $\text{rank}(\hat{\mathbf{B}}) \leq N$. Define the operator

$$\tilde{\mathbf{A}} := \mathbf{U}^\top \hat{\mathbf{B}} \mathbf{U} = \mathbf{U}^\top \mathbf{Y}' \mathbf{V} \mathbf{\Sigma}^{-1},$$

with eigendecomposition $\tilde{\mathbf{A}} \mathbf{W} = \mathbf{W} \mathbf{\Lambda}$ and set

$$\mathbf{\Phi} := \mathbf{Y}' \mathbf{V} \mathbf{\Sigma}^{-1} \mathbf{W}.$$

Columns of $\mathbf{\Phi}$ determine the DMD dynamic modes, so we call its column space the mode space. The following proposition summarizes key properties of $\hat{\mathbf{B}}$ and $\mathbf{\Phi}$.

Proposition 2. *Let $\mathbf{U}, \mathbf{\Sigma}, \mathbf{V}$ be as in (12) and let $\mathbf{P}_\Phi := \mathbf{\Phi} \mathbf{\Phi}^+$ denote the orthogonal projector from $\mathbb{R}^M$ onto the column space of $\mathbf{\Phi}$. Then the following hold:*

- (i) $\hat{\mathbf{B}} \mathbf{\Phi} = \mathbf{\Phi} \mathbf{\Lambda}$
- (ii) $\hat{\mathbf{B}} \mathbf{P}_\Phi = \mathbf{\Phi} \mathbf{\Lambda} \mathbf{\Phi}^+.$

Proof. See Appendix C.5.2. □

Proposition 2 implies that on the estimated mode space, $\hat{\mathbf{B}} = \hat{\mathbf{B}} \mathbf{P}_\Phi = \mathbf{\Phi} \mathbf{\Lambda} \mathbf{\Phi}^+$. Consequently, DMD provides a rank- N estimator $\hat{\mathbf{B}}$ of the first-order VAR coefficient matrix with spectral information about $\hat{\mathbf{B}}$ contained in eigenvectors $\mathbf{\Phi}$ and eigenvalues $\mathbf{\Lambda}$. The column space of $\mathbf{\Phi}$ estimates an invariant subspace of the coefficient matrix $\mathbf{B}^*$ in our VAR representation (6) and the corresponding eigenvalues in $\mathbf{\Lambda}$ estimate the associated nonzero eigenvalues of $\mathbf{B}^*$. When $\text{rank}(\mathbf{G}) = N_x$ and $\mathbf{\Sigma}_x \succ 0$, these nonzero eigenvalues equal the eigenvalues of the underlying state transition matrix $\mathbf{A}$.

Algorithm 1 summarizes the exact DMD procedure for estimating the reduced-rank VAR operator $\hat{\mathbf{B}}$, DMD modes Φ , and eigenvalues Λ from an $M \times (T + 1)$ data matrix $\mathbf{Y}$.

Algorithm 1 Exact DMD for reduced-rank VAR estimation (Tu et al., 2014)

Input: Centered data matrix $[\mathbf{y}_1, \dots, \mathbf{y}_{T+1}]$ and truncation rank N

1. Form data matrices:

$$\mathbf{Y} = [\mathbf{y}_1, \dots, \mathbf{y}_T], \quad \mathbf{Y}' = [\mathbf{y}_2, \dots, \mathbf{y}_{T+1}]$$

2. Compute rank- N truncated SVD of $\mathbf{Y}$:

$$\mathbf{Y} \approx \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top$$

3. Compute least-squares estimator:

$$\hat{\mathbf{B}} = \mathbf{Y}' \mathbf{V} \mathbf{\Sigma}^{-1} \mathbf{U}^\top$$

4. Compute reduced-rank dynamics:

$$\tilde{\mathbf{A}} = \mathbf{U}^\top \hat{\mathbf{B}} \mathbf{U} = \mathbf{U}^\top \mathbf{Y}' \mathbf{V} \mathbf{\Sigma}^{-1}$$

5. Compute DMD modes and eigenvalues:

$$\tilde{\mathbf{A}} \mathbf{W} = \mathbf{W} \Lambda$$

$$\Phi = \mathbf{Y}' \mathbf{V} \mathbf{\Sigma}^{-1} \mathbf{W}$$

6. Construct spectral DMD operator:

$$\tilde{\mathbf{B}} = \Phi \Lambda \Phi^+$$

Output: $\hat{\mathbf{B}}, \Phi, \Lambda, \tilde{\mathbf{B}}$

2.2 State-Space Representation of DMD

DMD delivers both a least-squares coefficient matrix $\hat{\mathbf{B}}$ and a rank- N spectral operator $\tilde{\mathbf{B}} := \Phi \Lambda \Phi^+$. By Proposition 2, $\tilde{\mathbf{B}} = \hat{\mathbf{B}} \mathbf{P}_\Phi$, so $\hat{\mathbf{B}}$ and $\tilde{\mathbf{B}}$ coincide on the estimated mode

space. To focus our exposition on links to the state-space representations (1)–(3), we now assume that $N = N_x$ and that $\text{rank}(\mathbf{G}) = N_x$ so that the measurement map is injective. In this case, Proposition 1 implies that $\{\mathbf{y}_t\}$ is itself a rank- N_x VAR(1), so that the pseudo-innovations representation (17) below aligns with representation (3). Assume that Φ has full column rank and define $\tilde{\mathbf{x}}_t := \Phi^+ \mathbf{y}_t$ and residuals

$$\tilde{\mathbf{a}}_{t+1} := \mathbf{y}_{t+1} - \tilde{\mathbf{B}} \mathbf{y}_t = \mathbf{y}_{t+1} - \Phi \Lambda \tilde{\mathbf{x}}_t.$$

Define the associated residual covariance estimator

$$\tilde{\Omega} := \frac{1}{T} \sum_{t=2}^{T+1} \tilde{\mathbf{a}}_t \tilde{\mathbf{a}}_t^\top.$$

Then the DMD estimated rank- N first-order VAR has the state-space representation

$$\begin{aligned} \tilde{\mathbf{x}}_{t+1} &= \Lambda \tilde{\mathbf{x}}_t + \Phi^+ \tilde{\mathbf{a}}_{t+1}, \\ \mathbf{y}_{t+1} &= \Phi \Lambda \tilde{\mathbf{x}}_t + \tilde{\mathbf{a}}_{t+1}. \end{aligned} \tag{15}$$

Under a maintained hypothesis (1) that includes $N = N_x$ and $\text{rank}(\mathbf{G}) = N_x$, the $\{\mathbf{y}_t\}$ process is a VAR(1), so in the population version of (15) the prediction errors are the VAR innovations and satisfy⁹

$$\mathbb{E} \left[\tilde{\mathbf{a}}_{t+1} \mathbf{y}_{t-j}^\top \right] = \mathbf{0} \quad \text{for all } j \geq 0.$$

2.3 Pseudo Innovations Representation

We now recast (15) as a pseudo-innovations representation by introducing a one-step-ahead predictor state for the fitted reduced-rank VAR(1). We maintain the full-

⁹Appendix C.3 extends our analysis beyond the assumptions in this section. It describes the associated least-squares reduced-coordinate representation and the associated normal-equation orthogonality (Proposition 4). Appendix C.4 studies the general mode-space representation for $\tilde{\mathbf{B}}$ (Corollary 2).

column-rank condition on Φ so that (15) applies. Recycling our $\hat{\cdot}$ notation, define the one-step-ahead least-squares predictor

$$\hat{\mathbf{x}}_t := \Lambda \tilde{\mathbf{x}}_{t-1} = \Lambda \Phi^+ \mathbf{y}_{t-1}. \quad (16)$$

We ask the reader to forgive us for recycling the $\hat{\mathbf{x}}_t$. Here we have constructed it from the fitted VAR(1), whereas earlier it referred to $\hat{\mathbf{x}}_t$ in (3).

Using the measurement equation in (15) at time $t - 1$, we have $\mathbf{y}_t = \Phi \hat{\mathbf{x}}_t + \tilde{\mathbf{a}}_t$. Multiplying the state equation in (15) at time $t - 1$ by Λ gives $\hat{\mathbf{x}}_{t+1} = \Lambda \hat{\mathbf{x}}_t + \Lambda \Phi^+ \tilde{\mathbf{a}}_t$. We thus have the following pseudo-innovations representation

$$\begin{aligned} \hat{\mathbf{x}}_{t+1} &= \Lambda \hat{\mathbf{x}}_t + \Lambda \Phi^+ \tilde{\mathbf{a}}_t, \\ \mathbf{y}_t &= \Phi \hat{\mathbf{x}}_t + \tilde{\mathbf{a}}_t. \end{aligned} \quad (17)$$

Equations (15) and (17) show how DMD estimates yield a state-space representation of a reduced rank- N VAR(1). When $\text{rank}(\mathbf{G}) = N_x$ and $N = N_x$ hold, in population this fitted representation coincides with the innovations representation (3) from Section 2 up to a similarity transformation of the state. This means that we can estimate the key matrices appearing in the state-space representations (1)–(3) of our maintained hypothesis by setting

$$\hat{\mathbf{A}} := \Lambda, \quad \hat{\mathbf{K}} := \Lambda \Phi^+, \quad \hat{\mathbf{G}} := \Phi. \quad (18)$$

In Section 3, we shall use information-projection to study how DMD estimates of a “Kepler stage” Section 2 model with equivalent state space representations (1) and (3) can be interpreted in terms of a “Newton-stage” structural dynamic stochastic general equilibrium model of a type widely used to study insurance and redistribution.

3 Information Projection of a Structural Model onto DMD

In the spirit of [Koopmans \(1947\)](#) and [Sargent and Sims \(1977\)](#), we regard our Section 2 model as a Kepler-stage descriptive model that can be used to detect and organize data patterns that a Newton-stage structural model should fit. [Lucas \(1987\)](#) and other leading 20th-century macroeconomists viewed descriptive findings like those of [Burns and Mitchell \(1946\)](#) as providing the empirical foundation for a “neo-classical synthesis” that separates macroeconomic analysis of aggregate business cycles and growth from microeconomic analysis of redistribution and social insurance.¹⁰ That descriptive statistical work indicates that the dynamics of aggregate quantities reported in the National Income and Product Accounts emerged from the operation of a “homogenizing mechanism” that aggregates a myriad of microeconomic shocks into one or two macroeconomic “factors”.¹¹ According to [Koopmans \(1947\)](#) and [Lucas \(1987\)](#), the role of a structural macroeconomic model is to interpret and explain these regularities in terms of parameters that describe preferences, technologies, and information flows.¹²

This section uses as a homogenizing mechanism a competitive equilibrium in the complete-markets tradition of Arrow–Debreu. The model’s equilibrium includes a vector stochastic process for a cross-section of households’ income and consumption that we then describe with our Section 2 statistical model. In this way, we relate our DMD-recovered parameters to parameters of the underlying structural model. This lab experiment is designed partly to help us understand an application of DMD to actual US data, namely, [Sargent et al. \(2025\)](#). That paper uses our Section 2 model to study the dynamics of CEX cross-sections of US personal income and consumption. To interpret outcomes in that paper, the off-the-shelf Arrow–Debreu model employed

¹⁰See [Sargent \(2015, 2024\)](#).

¹¹Milton Friedman suggested the term “homogenizing mechanism” to Sargent in a personal conversation in 1976.

¹²[Sargent \(2022, pp. 22-23\)](#) describes unpublished comments on [Sargent and Sims \(1977\)](#) by Robert E. Lucas, Jr. in which Lucas used the structural model presented in [Lucas \(1975\)](#) to interpret [Sargent and Sims](#)’s finding of two dynamic factors in their frequency domain analysis. [Sargent](#) tells how repeated post-conference revisions by [Sargent and Sims](#) caused Lucas to decline to publish his comments in the conference volume.

here is a convenient choice. It is a member of a class of readily computable linear-quadratic structures presented by [Hansen and Sargent \(2013\)](#). Thus, in the spirit of [White \(1982\)](#), [Hansen and Sargent \(1993\)](#), and [Fernández-Villaverde and Rubio-Ramírez \(2004\)](#), our purpose here is to study properties of our Section 2 descriptive model as a presumably misspecified approximation to an interesting structural model. We do this by constructing an “information projection” of the structural model onto the DMD model.¹³

[Sargent et al. \(2025\)](#) detected evidence of substantial redistribution and insurance in US CEX data from 1990 to 2023. The structural model presented in this section delivers such outcomes. It is a linear-quadratic DSGE economy with heterogeneous households whose preferences satisfy Gorman aggregation conditions ([Hansen and Sargent, 2013](#), ch. 12). Competitive equilibrium prices and aggregate quantities can be computed without tracking the cross-sectional wealth distribution. Time-invariant Pareto weights determine how household-level consumption allocations respond to aggregate dynamics.

In our structural model, two “homogenizing mechanisms” are active, one exogenous, the other endogenous. We build in the exogenous part of the mechanism when we specify households’ stochastic consumption endowments. We assume that the flows of incomes from households’ “Lucas trees” share a common dynamic factor.¹⁴ The endogenous part of the homogenizing mechanism comes from the extensive risk-sharing that competitive and complete markets deliver. It is easy for us to calculate efficient redistribution schemes by appropriately tilting a vector of implied Pareto weights that emerge from an initial competitive equilibrium allocation. We take advantage of that feature to generate outcomes that resemble ones actually uncovered by the statistical analysis with the CEX data in [Sargent et al. \(2025\)](#), as exhibited by comparing Figure 3

¹³See footnote 18 below for an explanation of “information projection”. See [Sargent \(1976\)](#) for an early application.

¹⁴[Eberly and Wang \(2025\)](#) provide the canonical “two-tree” model that extends the classic one-tree model of [Lucas \(1978\)](#). Our [Hansen and Sargent \(2013, ch. 12\)](#) model has as many trees as households.

below with Figure 3 in [Sargent et al. \(2025\)](#). Among other things, we study how well DMD recovers low-dimensional aggregate dynamics that drive household incomes and consumptions in our DSGE model.

3.1 The Structural DSGE Model

Our model is a version of one described by [Hansen and Sargent \(2013, ch. 12\)](#). Time is discrete, $t \in \{0, 1, 2, \dots\}$. There are J households, indexed by j , who share a common discount factor $\beta \in (0, 1)$ and information set, but differ in preference shocks and endowments. Households consume a single final good and supply a single intermediate input. Household j chooses $\{c_{jt}, \ell_{jt}\}_{t \geq 0}$ to maximize

$$-\frac{1}{2} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[(s_{jt} - b_{jt})^\top (s_{jt} - b_{jt}) + \ell_{jt}^\top \ell_{jt} \right] \quad (19)$$

subject to a household service technology

$$s_{jt} = \Lambda_s h_{j,t-1} + \Pi_s c_{jt},$$

$$h_{jt} = \Delta_h h_{j,t-1} + \Theta_h c_{jt},$$

and an Arrow–Debreu time-0 intertemporal budget constraint

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t p_{0t} \cdot c_{jt} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t (w_{0t} \ell_{jt} + \alpha_{0t} \cdot d_{jt}) + v_0 \cdot k_{j,-1},$$

Here c_{jt} is consumption, ℓ_{jt} is labor supply, h_{jt} is a household durable stock, and s_{jt} is the associated service flow. p_{0t} is the price of consumption, w_{0t} is the price of the intermediate input, α_{0t} is the price vector of endowment goods, and v_0 is the price of the initial capital stock $k_{j,-1}$. Bliss points $b_{jt} = \mathbf{U}_b^j \mathbf{z}_t$ and endowments $d_{jt} = \mathbf{U}_d^j \mathbf{z}_t$ are

linear in an exogenous state vector $\mathbf{z}_t \in \mathbb{R}^{n_z}$ that follows

$$\mathbf{z}_{t+1} = \mathbf{A}_{22} \mathbf{z}_t + \mathbf{C}_2 \mathbf{w}_{t+1}, \quad \mathbf{w}_{t+1} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (20)$$

Heterogeneity enters only through the loadings $(\mathbf{U}_b^j, \mathbf{U}_d^j)$ and initial stocks $(h_{j,-1}, k_{j,-1})$; the technology parameters $(\Lambda_s, \Pi_s, \Delta_h, \Theta_h)$ are common across households. This structure implies that individual demands are affine in wealth with common marginal propensities, a necessary condition for Gorman aggregation ([Gorman, 1953](#)).

3.2 Aggregate Dynamics and the Sharing Rule

Competitive equilibrium allocations are Pareto efficient and can be computed by solving a representative-agent planning problem. Define aggregates $c_t := \sum_j c_{jt}$, $h_t := \sum_j h_{jt}$, $k_t := \sum_j k_{jt}$, and let $\mathbf{U}_b := \sum_j \mathbf{U}_b^j$, $\mathbf{U}_d := \sum_j \mathbf{U}_d^j$ so that $b_t = \mathbf{U}_b \mathbf{z}_t$ and $d_t = \mathbf{U}_d \mathbf{z}_t$. The planner maximizes

$$-\frac{1}{2} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[(s_t - b_t)^\top (s_t - b_t) + g_t^\top g_t \right] \quad (21)$$

subject to

$$\begin{aligned} \Psi_c c_t + \Psi_g g_t + \Psi_i i_t &= \Gamma k_{t-1} + d_t, \quad k_t = \Delta_k k_{t-1} + \Theta_k i_t, \\ h_t &= \Delta_h h_{t-1} + \Theta_h c_t, \quad s_t = \Lambda_s h_{t-1} + \Pi_s c_t. \end{aligned}$$

In (21), g_t is the aggregate intermediate good and i_t is investment. Let μ_{0j}^w denote household j 's time-zero marginal utility of wealth and define the implied Pareto weight associated with the competitive equilibrium for an economy with initial stocks $(k_{j,-1}, h_{j,-1})$ and exogenous endowments $\{d_{j,t}\}$ across agents j as

$$\mu_j := \frac{\mu_{0j}^w}{\sum_{i=1}^J \mu_{0i}^w}. \quad (22)$$

The consumption allocation rule takes the form

$$c_{jt} = \mu_j c_t + \tilde{\chi}_{jt}, \quad (23)$$

where $\mu_j c_t$ is household j 's proportional share of aggregate consumption and $\tilde{\chi}_{jt}$ is a deviation term determined by preference heterogeneity associated with idiosyncratic preference shocks. The deviations sum to zero: $\sum_j \tilde{\chi}_{jt} = 0$. The implied Pareto weight μ_j depends only on time-0 conditions and is time-invariant. Holding aggregate endowments and initial aggregate stocks fixed, if we assign a new set of Pareto weights that sum to 1, then (23) describes a new allocation that leaves aggregate dynamics unchanged. In this way, we can describe the outcomes of a Pareto-efficient redistribution scheme while leaving unspecified the detailed timing and state-dependencies that produce it. There is an equivalence class of distributions of initial capital stocks and endowments that validate a given set of Pareto weights. We therefore redistribute by simply reassigning Pareto weights.¹⁵

3.3 A 100-Household Example

We now construct a large-panel experiment in which DMD is applied to a high-dimensional cross-section of household outcomes generated by the DSGE model described in Section 3.1. In our application, we muted idiosyncratic preference shocks, so the deviation term $\tilde{\chi}_{jt}$ in (23) is zero. Set $J = 100$ and define vectors $\boldsymbol{\eta}_t := (\eta_{J_a+1,t}, \dots, \eta_{J,t})'$ as the idiosyncratic endowment states for $j > J_a$. The exoge-

¹⁵For a version of our model in which preference shocks are shut down, an Arrow–Debreu complete-markets allocation can be implemented with a limited set of markets: a mutual fund holding all endowment claims and a one-period riskless bond. When the riskless return is constant and the deviation baseline $\{\tilde{\chi}_{jt}\}$ is known at time 0, all households hold the same portfolio of risky assets (the mutual fund), and dynamic rebalancing occurs solely through the bond market. This trading structure implements an efficient allocation and extends a two-fund theorem of Rubinstein (1974) to a multiperiod setting. See Hansen and Sargent (2013, Sect. 12.6). Sargent and Stachurski (2025) provide more details.

nous state is

$$\mathbf{z}_t = \begin{bmatrix} 1 \\ d_{a,t} \\ d_{a,t-1} \\ \boldsymbol{\eta}_t \end{bmatrix}.$$

The aggregate component follows the AR(1) process

$$d_{a,t+1} = \rho_1 d_{a,t} + \sigma_a \boldsymbol{\varepsilon}_{a,t+1}, \quad (\rho_1, \sigma_a) = (0.95, 0.5).$$

where $\boldsymbol{\varepsilon}_{a,t+1} \sim \mathcal{N}(0, 1)$ is the aggregate innovation (a component of $\mathbf{w}_{t+1}$ in (20)). We set $J_a = 50$ and construct endowments so that idiosyncratic shocks sum to zero:

$$\begin{aligned} d_{jt} &= \alpha_j + \phi_j d_{a,t} + \eta_{j,t}, \quad j = J_a + 1, \dots, J, \\ d_{jt} &= \alpha_j + \phi_j d_{a,t} - \frac{1}{J_a} \sum_{k=J_a+1}^J \eta_{k,t}, \quad j = 1, \dots, J_a. \end{aligned}$$

In this setting, $\sum_{j=1}^J d_{jt}$ depends on $\{d_{a,t}\}$ but not on $\{\eta_{j,t}\}$ once we normalize $\sum_{j=1}^J \phi_j = 1$. We draw $\alpha_j \sim \mathcal{U}[3, 5]$ and $\tilde{\phi}_j \sim \mathcal{U}[0.5, 1.5]$, then set $\phi_j = \tilde{\phi}_j / \sum_{i=1}^J \tilde{\phi}_i$. To generate an economy where low-income households experience more idiosyncratic risk, we define p_j as one minus the percentile rank of α_j in the cross-section and set, for $j > J_a$,

$$\sigma_j = 0.2 + (5.0 - 0.2) p_j^2, \quad \rho_j^d = 0.0 + (0.90 - 0.0) p_j,$$

in the AR(1) process $\eta_{j,t+1} = \rho_j^d \eta_{j,t} + \sigma_j \boldsymbol{\varepsilon}_{j,t+1}^d$. The innovations $\{\boldsymbol{\varepsilon}_{j,t+1}^d\}$ are i.i.d. $\mathcal{N}(0, 1)$ across j and t and are components of $\mathbf{w}_{t+1}$. We interpret changes in Pareto weights as an incompletely described, Pareto-efficient tax-and-transfer scheme that redistributes resources across households without changing aggregate dynamics. To implement redistribution, we compute an alternative Pareto-efficient allocation that replaces the

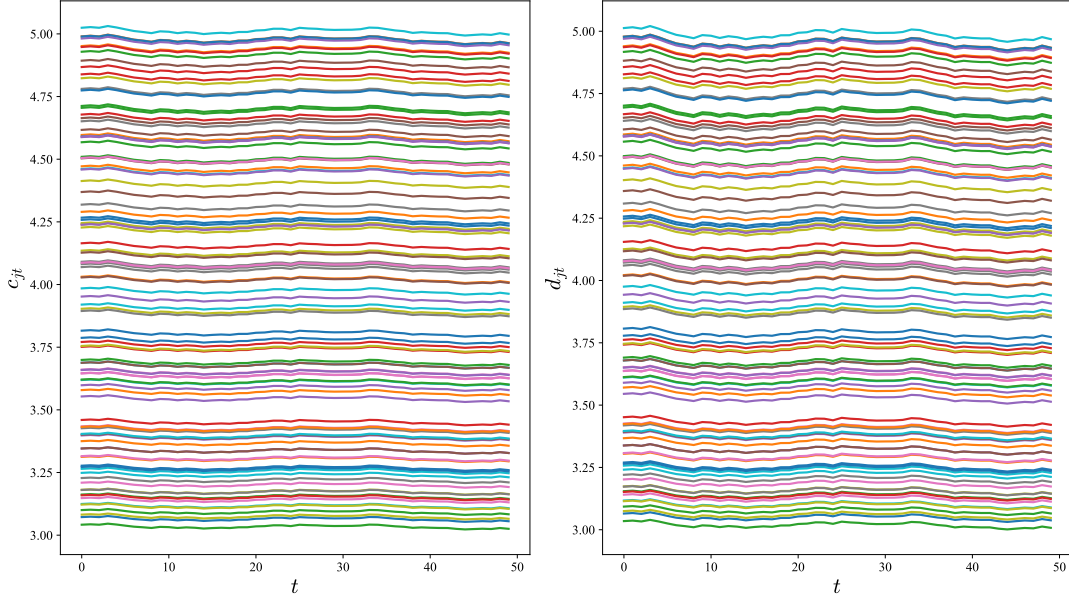


Figure 1: Household consumption and post-trade dividend-income paths in the 100-household economy (after burn-in). Each line corresponds to a household.

competitive equilibrium Pareto weights $\{\mu_j\}$ with new Pareto weights $\{\mu_j^*\}$ defined by

$$\mu_j^* = (1 - a)\mu_j + a/J, \quad a = 0.8,$$

so that $\sum_{j=1}^J \mu_j^* = 1$. As mentioned above, there is an equivalence class of efficient tax-and-transfer systems that imply these Pareto weights.

We simulate our DSGE model for 2,000 periods and discard the first 200 as burn-in. Figure 1 plots simulated household consumption and post-trade dividend-income panels, and Figure 2 summarizes the post-redistribution Pareto weights. Under the sharing rule (23), the aggregate component of consumption is $\mu_j c_t$, the source of strong comovements across households (Figure 1).

Figure 2 shows how our redistribution tilts Pareto weights away from those associated with our initial competitive equilibrium toward more equality. Thanks to conditions for Gorman aggregation being satisfied, redistribution rearranges the cross-

sectional allocation but leaves the aggregate dynamics of $\{c_t, k_t\}$ unchanged.

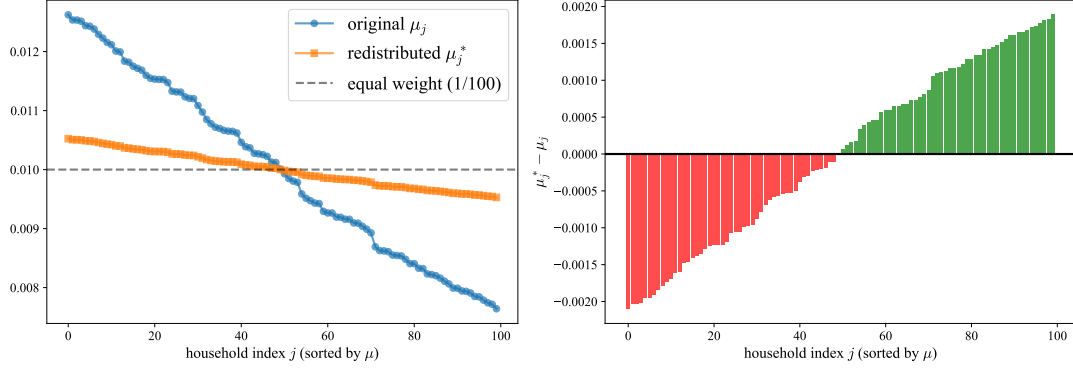


Figure 2: Pareto weights in the 100-household economy. The left panel plots the Pareto weights $\{\mu_j\}$ and redistributed weights $\{\mu_j^*\}$ after sorting households by μ_j . The right panel plots $\mu_j^* - \mu_j$.

To visualize redistribution, we define a household income process and study how redistribution affects household incomes and consumptions. Let $y_{jt}(\omega)$ be household j 's net income for a given Pareto-weight vector ω , and define household j 's net income at time t as

$$y_{jt}(\omega) = \omega_j d_t + (R - 1)w_{j,t-1}, \quad (24)$$

where d_t is the aggregate endowment, $R = 1/\beta$ is the gross return, and $w_{j,t-1} = \omega_j k_{t-1} + \hat{k}_{j,t-1}$ is household j 's total asset comprising a proportional capital share $\omega_j k_{t-1}$ and a bond position $\hat{k}_{j,t-1}$. The first term is dividend income from holding ω_j shares of the aggregate endowment; the second term is the net return on wealth. Let $y_{jt}^{pre} := y_{jt}(\mu)$ denote household j 's net income under the competitive equilibrium weight vector $\{\mu_j\}$, let $y_{jt}^{post} := y_{jt}(\mu^*)$ denote net income under the redistributed weights $\{\mu_j^*\}$, and let c_{jt}^{post} denote post-redistribution consumption.

Effects of redistribution on household incomes and consumptions are depicted in Figure 3, which plots cross-sectional percentiles of pre- and post-redistribution income and post-redistribution consumption. Since household shares are time-invariant in this

experiment, each percentile corresponds to the same households over time.

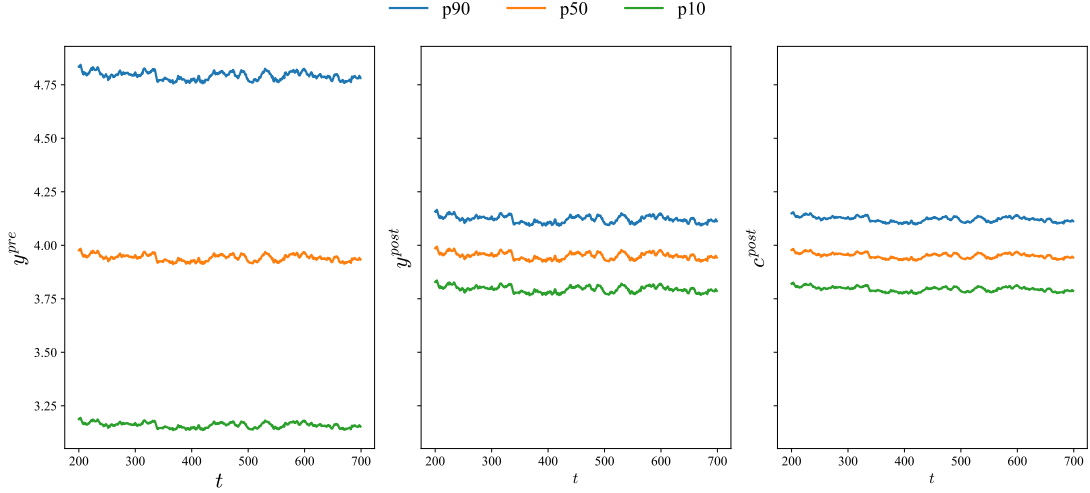


Figure 3: Cross-sectional percentiles (p10, p50, p90) of pre-redistribution income y_{jt}^{pre} , post-redistribution income y_{jt}^{post} , and post-redistribution consumption c_{jt}^{post} in the 100-household experiment (after burn-in).

3.4 Ground Truth State-Space Representation

A solution to the planner's problem yields equilibrium prices and allocations as linear functions of the aggregate state and a state-space representation for a high-dimensional household panel. The aggregate state vector is

$$\mathbf{x}_t = \begin{bmatrix} h_{t-1} \\ k_{t-1} \\ \mathbf{z}_t \end{bmatrix},$$

where h_{t-1} is the aggregate household durable stock and k_{t-1} is the aggregate capital stock. The equilibrium law of motion for the state evolves according to

$$\mathbf{x}_{t+1} = \mathbf{A}_o \mathbf{x}_t + \mathbf{C}_o \mathbf{w}_{t+1}, \quad \mathbf{w}_{t+1} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (25)$$

Each equilibrium quantity is a linear function of the state. Stacking aggregate consumption, the intermediate input, investment, household durables, capital, endowments, preference bliss points, and services gives a measurement equation

$$\mathbf{y}_t^{\text{agg}} := \begin{bmatrix} c_t \\ g_t \\ i_t \\ h_t \\ k_t \\ d_t \\ b_t \\ s_t \end{bmatrix} = \mathbf{G}_o^{\text{agg}} \mathbf{x}_t, \quad (26)$$

where $\mathbf{G}_o^{\text{agg}}$ stacks pertinent selection vectors.

Suppose that an econometrician observes only household-level income and consumption data. The econometrician observes a high-dimensional panel that stacks households' pre- and post-redistribution incomes and post-redistribution consumptions as

$$\mathbf{y}_t^{\text{GL}} := \begin{bmatrix} \mathbf{y}_t^{\text{pre}} \\ \mathbf{y}_t^{\text{post}} \\ \mathbf{c}_t^{\text{post}} \end{bmatrix} \in \mathbb{R}^M, \quad M = 3J,$$

where $\mathbf{y}_t^{\text{pre}} := (y_{1t}^{\text{pre}}, \dots, y_{Jt}^{\text{pre}})^\top$, $\mathbf{y}_t^{\text{post}} := (y_{1t}^{\text{post}}, \dots, y_{Jt}^{\text{post}})^\top$, and $\mathbf{c}_t^{\text{post}} := (c_{1t}^{\text{post}}, \dots, c_{Jt}^{\text{post}})^\top$ are J -vectors. With idiosyncratic preference shocks muted, each component of $\mathbf{y}_t^{\text{GL}}$ is a linear function of the centered state (see Appendix B), so the household panel has a state-space representation of the form

$$\begin{aligned} \mathbf{x}_{t+1} &= \mathbf{A}_o \mathbf{x}_t + \mathbf{C}_o \mathbf{w}_{t+1}, \\ \mathbf{y}_t^{\text{GL}} &= \mathbf{G}_o \mathbf{x}_t, \end{aligned} \quad (27)$$

which is a special case of (1) with $(\mathbf{A}, \mathbf{C}, \mathbf{G}) = (\mathbf{A}_o, \mathbf{C}_o, \mathbf{G}_o)$. The measurement matrix $\mathbf{G}_o$ has the block structure

$$\mathbf{G}_o = \begin{bmatrix} \mathbf{G}^{y,\text{pre}} \\ \mathbf{G}^{y,\text{post}} \\ \mathbf{G}^{c,\text{post}} \end{bmatrix} \quad \mathbf{G}_o \in \mathbb{R}^{M \times N_x},$$

where $N_x = 54$ is the dimension of $\mathbf{x}_t$.¹⁶ and $\mathbf{G}^{y,\text{pre}}, \mathbf{G}^{y,\text{post}}, \mathbf{G}^{c,\text{post}} \in \mathbb{R}^{J \times N_x}$ map $\mathbf{x}_t$ to $\mathbf{y}_t^{\text{pre}}, \mathbf{y}_t^{\text{post}}$, and $\mathbf{c}_t^{\text{post}}$, respectively. Appendix B derives formulas for the blocks of $\mathbf{G}_o$.

We study a $T = 250$ subsample of the post-burn-in data. Here, $M = 300$ and $T = 250$, so $M > T$.¹⁷ The household panel $\{\mathbf{y}_t^{\text{GL}}\}_{t=1}^T$ is our input to the DMD procedure of Section 2.1 to estimate a reduced-rank VAR operator $\hat{\mathbf{B}}$ and its eigendecomposition. In our laboratory, we generate aggregates from the linear state-space system (25)–(26), but the econometrician observes only microdata at the household level $\{\mathbf{y}_t^{\text{GL}}\}_{t=1}^T$. We assess DMD’s performance through diagnostics that compare estimated objects to their population counterparts. After describing rank selection and state recovery, we discuss cross-sectional loadings and forecasting.

Before examining empirical results, it is useful to record the population object that DMD targets. Since our DMD implementation is applied to a demeaned data matrix, we work with the centered panel $\mathbf{y}_t := \mathbf{y}_t^{\text{GL}} - \bar{\mathbf{y}}$, where $\bar{\mathbf{y}}$ is the sample mean of $\{\mathbf{y}_t^{\text{GL}}\}_{t=1}^T$. The centered panel $\mathbf{y}_t$ admits a state-space representation that parallels (1)–(3). Let $(\mathbf{A}_{\text{GL}}, \mathbf{C}_{\text{GL}}, \mathbf{G}_{\text{GL}})$ denote the state transition, shock loading, and measurement matrices of the centered state-space system (i.e., with the constant state component

¹⁶The 54 states consist of $h_{t-1}, k_{t-1}, d_{a,t}, d_{a,t-1}$, and $\boldsymbol{\eta}_t \in \mathbb{R}^{J-J_a}$ with $J - J_a = 50$.

¹⁷We simulate 2,000 periods, discard the first 200 periods as burn-in, and extract a 250-period subsample for DMD analysis.

removed from $(\mathbf{A}_o, \mathbf{C}_o, \mathbf{G}_o)$. The centered state-space representation is

$$\begin{aligned}\mathbf{x}_{t+1} &= \mathbf{A}_{\text{GL}} \mathbf{x}_t + \mathbf{C}_{\text{GL}} \mathbf{w}_{t+1}, \\ \mathbf{y}_t &= \mathbf{G}_{\text{GL}} \mathbf{x}_t,\end{aligned}\tag{28}$$

where $\mathbf{w}_{t+1} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The associated steady-state innovations representation is

$$\begin{aligned}\hat{\mathbf{x}}_{t+1} &= \mathbf{A}_{\text{GL}} \hat{\mathbf{x}}_t + \mathbf{K}_{\text{GL}} \mathbf{a}_t, \\ \mathbf{y}_t &= \mathbf{G}_{\text{GL}} \hat{\mathbf{x}}_t + \mathbf{a}_t,\end{aligned}\tag{29}$$

where $\hat{\mathbf{x}}_t := \mathbb{E}[\mathbf{x}_t \mid \mathbf{y}^{t-1}]$ is the one-step-ahead state prediction, $\mathbf{a}_t := \mathbf{y}_t - \mathbf{G}_{\text{GL}} \hat{\mathbf{x}}_t$ is the innovation, and the steady-state Kalman gain is

$$\mathbf{K}_{\text{GL}} := \mathbf{A}_{\text{GL}} \boldsymbol{\Sigma}_{\infty} \mathbf{G}_{\text{GL}}^{\top} (\mathbf{G}_{\text{GL}} \boldsymbol{\Sigma}_{\infty} \mathbf{G}_{\text{GL}}^{\top})^+, \tag{30}$$

with $\boldsymbol{\Sigma}_{\infty} := \mathbb{E}[(\mathbf{x}_t - \hat{\mathbf{x}}_t)(\mathbf{x}_t - \hat{\mathbf{x}}_t)^{\top}]$ denoting the steady-state prediction error covariance.

To derive the vector autoregressive representation, iterate (29) forward. From the first equation, $\hat{\mathbf{x}}_t = \mathbf{A}_{\text{GL}} \hat{\mathbf{x}}_{t-1} + \mathbf{K}_{\text{GL}} \mathbf{a}_{t-1}$. Substituting into the second equation of (29) gives

$$\mathbf{y}_t = \mathbf{G}_{\text{GL}} \mathbf{A}_{\text{GL}} \hat{\mathbf{x}}_{t-1} + \mathbf{G}_{\text{GL}} \mathbf{K}_{\text{GL}} \mathbf{a}_{t-1} + \mathbf{a}_t. \tag{31}$$

From the second equation of (29) at $t-1$, we have $\mathbf{G}_{\text{GL}} \hat{\mathbf{x}}_{t-1} = \mathbf{y}_{t-1} - \mathbf{a}_{t-1}$. Repeated substitution yields

$$\mathbf{y}_t = \sum_{j=1}^{\infty} \mathbf{G}_{\text{GL}} (\mathbf{A}_{\text{GL}} - \mathbf{K}_{\text{GL}} \mathbf{G}_{\text{GL}})^{j-1} \mathbf{K}_{\text{GL}} \mathbf{y}_{t-j} + \mathbf{a}_t. \tag{32}$$

Define the VAR coefficient matrices

$$\mathbf{B}_j := \mathbf{G}_{\text{GL}} (\mathbf{A}_{\text{GL}} - \mathbf{K}_{\text{GL}} \mathbf{G}_{\text{GL}})^{j-1} \mathbf{K}_{\text{GL}}, \quad j \geq 1 \tag{33}$$

Then the infinite-order VAR representation becomes

$$\mathbf{y}_t = \sum_{j=1}^{\infty} \mathbf{B}_j \mathbf{y}_{t-j} + \mathbf{a}_t. \quad (34)$$

This is a vector autoregressive representation for the centered household panel $\{\mathbf{y}_t\}$.

Let $\mathbf{\Omega}_{\mathbf{x}} := \text{Cov}(\mathbf{x}_t)$ denote the stationary covariance of the centered state. The population projection coefficient is the coefficient matrix in the linear projection of $\mathbf{y}_t$ on $\mathbf{y}_{t-1}$:

$$\mathbf{B}^* := \text{Cov}(\mathbf{y}_t, \mathbf{y}_{t-1}) \text{Cov}(\mathbf{y}_{t-1})^+ = \mathbf{G}_{\text{GL}} \mathbf{A}_{\text{GL}} \mathbf{\Omega}_{\mathbf{x}} \mathbf{G}_{\text{GL}}^{\top} \left(\mathbf{G}_{\text{GL}} \mathbf{\Omega}_{\mathbf{x}} \mathbf{G}_{\text{GL}}^{\top} \right)^+. \quad (35)$$

When $\mathbf{B}^*$ has (effective) rank at most N , a rank- N DMD estimator targets $\mathbf{B}^*$. We now examine the eigenvalue structure of $\mathbf{B}^*$ and $\{\mathbf{B}_j\}$ to assess the suitability of a low-rank approximation.

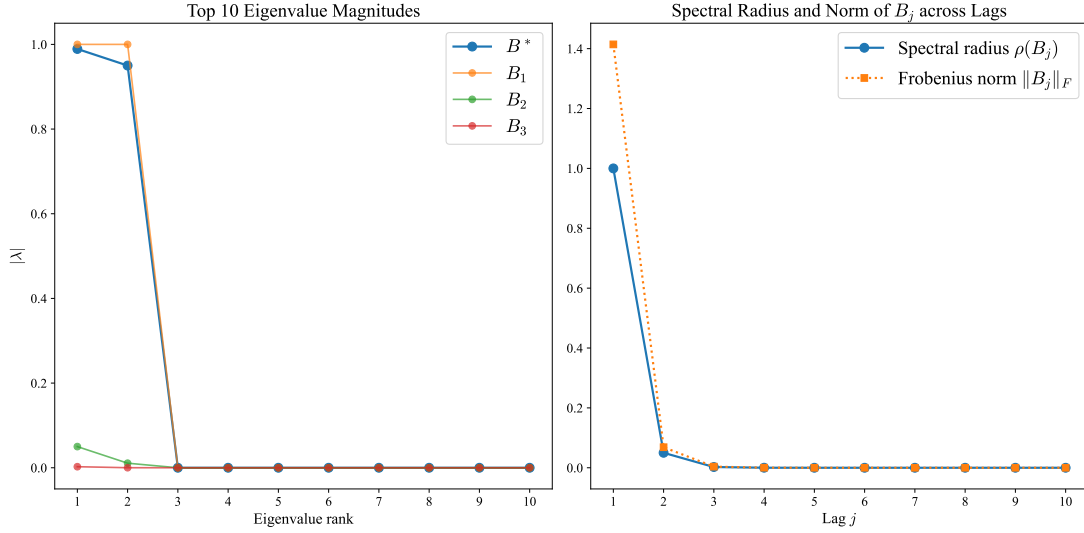


Figure 4: Eigenvalue structure of population projection and VAR coefficients. Left panel: Eigenvalue magnitudes of $\mathbf{B}^*$ and the first three infinite-order VAR coefficients $\mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_3$. Right panel: Spectral radius and Frobenius norm of $\mathbf{B}_j$ across lags.

Figure 4 provides evidence that $\mathbf{B}^*$ and the infinite-order VAR coefficients $\{\mathbf{B}_j\}$ are

both effectively low-rank. The left panel plots the eigenvalue magnitudes of $\mathbf{B}^*$ and the first three infinite-order VAR coefficients $\mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_3$ from (34). For each matrix, the top two eigenvalues dominate, while all other eigenvalues are numerically 0.

The nearly rank-2 structure of these projection and VAR coefficient matrices is inherited from the economic structure of our DSGE model. Under the sharing rule (23), household consumption $c_{jt} = \mu_j c_t + \tilde{\chi}_{jt}$ is dominated by the aggregate component $\mu_j c_t$, which under the planner's linear decision rule is itself a linear function of $(k_{t-1}, d_{a,t})$. Similarly, from the income definition (24), household income $y_{jt} = \omega_j d_t + (R-1)w_{j,t-1}$ depends on aggregate endowment d_t (a linear function of $d_{a,t}$) and wealth $w_{j,t-1}$, which loads on aggregate capital k_{t-1} . Although the state vector $\mathbf{x}_t$ contains idiosyncratic endowment states $\{\eta_{j,t}\}$ with dimension $J - J_a$, when idiosyncratic preference shocks are muted ($\tilde{\chi}_{jt} = 0$), the observables $\mathbf{y}_t^{\text{GL}}$ depend on the state only through aggregate factors (Appendix B). Hence, the centered panel lies in a two-dimensional subspace spanned by loadings on $(k_{t-1}, d_{a,t})$, and $\mathbf{B}^*$ inherits this low-rank structure.

The right panel of Figure 4 shows the spectral radius $\rho(\mathbf{B}_j)$ and Frobenius norm $\|\mathbf{B}_j\|_F$ for lags $j = 1, \dots, 10$. Both decay rapidly: the spectral radius falls from near unity at lag 1 to below 0.1 by lag 2, reflecting the geometric decay rate governed by eigenvalues of the closed-loop observer $\mathbf{A} - \mathbf{K}_{\text{GL}} \mathbf{G}$. This decay implies that even though the true representation (34) is an infinite-order VAR, most predictive information concentrates in the first lag, making a VAR(1) a good approximation. Heuristically, these results justify using a small N in DMD: when the spectrum of $\mathbf{B}^*$ is dominated by a few eigenvalues, a low-rank approximation captures most of the predictive content.

By calculating an “information projection” of our structural model onto the DMD family of statistical models, we can study the quality of our descriptive Section 2 model as an approximation to our structural DSGE.¹⁸ In the language of footnote

¹⁸Csiszár and Matus (2003) and Nielsen (2018) describe information projections. Let $\{f_\theta(x)\}_{\theta \in \Theta}$ and $\{g_\delta(x)\}_{\delta \in \Delta}$ be two collections (manifolds) of probability distributions for outcomes $x \in X$. When model $g_{\delta_o}(x)$ governs the data, a population maximum likelihood estimator θ_o of parameter vector $\theta \in \Theta$ of misspecified statistical model $f_\theta(x)$

18, let θ denote parameters of our DMD model and let δ denote parameters of our structural DSGE model. When the structural model with parameter δ_o generates the data, information projection selects the KL-minimizing DMD parameter θ_o . The mapping from the structural model's parameters δ_o to our DMD model's parameters θ_o tells us how the parameters of the approximating descriptive model reflect the parameters of the structural model. In the remainder of this section, we focus on this information projection.

To construct an information projection along lines described in Footnote 18 to our setting, let $p(\mathbf{y}_t \mid \mathbf{y}_{t-1})$ denote the DSGE model's conditional density and consider a Gaussian VAR(1) approximating family of models

$$q_{\mathbf{B}, \mathbf{\Omega}}(\mathbf{y}_t \mid \mathbf{y}_{t-1}) : \quad \mathbf{y}_t \mid \mathbf{y}_{t-1} \sim \mathcal{N}(\mathbf{B} \mathbf{y}_{t-1}, \mathbf{\Omega}), \quad \mathbf{\Omega} \succ \mathbf{0}.$$

The KL divergence of the DSGE model's conditional density from the approximating family of models is

$$\text{KL}(p, q_{\mathbf{B}, \mathbf{\Omega}}) := \mathbb{E} \left[\int \log \frac{p(\mathbf{y}_t \mid \mathbf{y}_{t-1})}{q_{\mathbf{B}, \mathbf{\Omega}}(\mathbf{y}_t \mid \mathbf{y}_{t-1})} p(\mathbf{y}_t \mid \mathbf{y}_{t-1}) d\mathbf{y}_t \right]. \quad (36)$$

A subtlety arises when $\mathbf{\Gamma}_0 := \mathbb{E}[\mathbf{y}_{t-1} \mathbf{y}_{t-1}^\top]$ is singular with $\text{rank}(\mathbf{\Gamma}_0) = r < M$. In this case, $\mathbf{y}_t$ is supported on an r -dimensional subspace $\mathcal{S} := \text{range}(\mathbf{\Gamma}_0)$, and the KL divergence (36) is infinite if our approximating model operates on $\mathbb{R}^M$ (since p is not absolutely continuous with respect to q). To obtain a well-defined projection, we restrict

minimizes the Kullback-Leibler divergence

$$\text{KL}(g_{\delta_o}, f_\theta) = \int \log \left(\frac{g_{\delta_o}(x)}{f_\theta(x)} \right) g_{\delta_o}(x) dx = -H(g_{\delta_o}) - \mathbb{E}_{g_{\delta_o}} \log f_\theta(x),$$

where $H(g_{\delta_o}) = \int \log \left(\frac{1}{g_{\delta_o}(x)} \right) g_{\delta_o}(x) dx$ is the Shannon information of nature's probability distribution $g_{\delta_o}(x)$ and $\mathbb{E}_{g_{\delta_o}}$ denotes a mathematical expectation under $g_{\delta_o}(x)$. The information projection of $g_{\delta_o}(x)$ onto $\{f_\theta(x)\}_{\theta \in \Theta}$ is distribution $f_{\theta_o}(x)$ in manifold $\{f_\theta(x)\}_{\theta \in \Theta}$ that maximum likelihood selects when nature's model g_{δ_o} generates the data, i.e., $\theta_o = \arg \max_{\theta \in \Theta} \mathbb{E}_{g_{\delta_o}} \log f_\theta(x)$ is the population maximum likelihood estimator of θ when probability distribution g_{δ_o} generates the data. [Fernández-Villaverde and Rubio-Ramírez \(2004\)](#) use information projections to study Bayesian estimation of misspecified structural dynamic equilibrium models.

it to the support $\mathcal{S}$. Let $\mathbf{R} \in \mathbb{R}^{M \times r}$ contain orthonormal basis vectors for $\mathcal{S}$ and define reduced coordinates $\tilde{\mathbf{y}}_t := \mathbf{R}^\top \mathbf{y}_t \in \mathbb{R}^r$. Since $\mathbf{y}_t \in \mathcal{S}$ almost surely, no information is lost and $\mathbf{y}_t = \mathbf{R}\tilde{\mathbf{y}}_t$.

The conditional KL divergence restricted to the support decomposes as

$$\text{KL}(p, q_{\mathbf{B}, \mathbf{\Omega}}) = \underbrace{\mathbb{E}[\log p(\tilde{\mathbf{y}}_t \mid \tilde{\mathbf{y}}_{t-1})]}_{-H(p)} - \mathbb{E}[\log q_{\mathbf{B}, \mathbf{\Omega}}(\tilde{\mathbf{y}}_t \mid \tilde{\mathbf{y}}_{t-1})], \quad (37)$$

where the conditional entropy $H(p)$ does not depend on $(\mathbf{B}, \mathbf{\Omega})$. Thus, minimizing KL divergence is equivalent to maximizing the expected log-likelihood.

Expanding the Gaussian log-density on the support gives

$$\begin{aligned} \text{KL}(p, q_{\mathbf{B}, \mathbf{\Omega}}) = & -H(p) + \frac{r}{2} \log(2\pi) + \frac{1}{2} \log \det(\tilde{\mathbf{\Omega}}) \\ & + \frac{1}{2} \mathbb{E}[(\tilde{\mathbf{y}}_t - \tilde{\mathbf{B}}\tilde{\mathbf{y}}_{t-1})^\top \tilde{\mathbf{\Omega}}^{-1} (\tilde{\mathbf{y}}_t - \tilde{\mathbf{B}}\tilde{\mathbf{y}}_{t-1})], \end{aligned} \quad (38)$$

where $\tilde{\mathbf{B}} := \mathbf{R}^\top \mathbf{B} \mathbf{R}$ and $\tilde{\mathbf{\Omega}} := \mathbf{R}^\top \mathbf{\Omega} \mathbf{R} \succ \mathbf{0}$. The first-order condition for minimizing over $\tilde{\mathbf{B}}$ is

$$\mathbb{E}[(\tilde{\mathbf{y}}_t - \tilde{\mathbf{B}}^* \tilde{\mathbf{y}}_{t-1}) \tilde{\mathbf{y}}_{t-1}^\top] = \mathbf{0}, \quad (39)$$

whose solution is $\tilde{\mathbf{B}}^* = \tilde{\mathbf{\Gamma}}_1 \tilde{\mathbf{\Gamma}}_0^{-1}$, where $\tilde{\mathbf{\Gamma}}_k := \mathbb{E}[\tilde{\mathbf{y}}_t \tilde{\mathbf{y}}_{t-k}^\top]$ and $\tilde{\mathbf{\Gamma}}_0 = \mathbf{R}^\top \mathbf{\Gamma}_0 \mathbf{R} \succ \mathbf{0}$.

Mapping back to $\mathbb{R}^M$ yields

$$\mathbf{B}^* := \mathbf{R} \tilde{\mathbf{B}}^* \mathbf{R}^\top = \mathbf{\Gamma}_1 \mathbf{\Gamma}_0^+,$$

where $\mathbf{\Gamma}_1 := \mathbb{E}[\mathbf{y}_t \mathbf{y}_{t-1}^\top]$ and $(\cdot)^+$ denotes the Moore–Penrose pseudoinverse. Thus, $\mathbf{B}^*$ is the information projection of the structural DSGE model's one-step-ahead conditional distribution onto the Gaussian VAR(1) family on subspace $\mathcal{S}$.

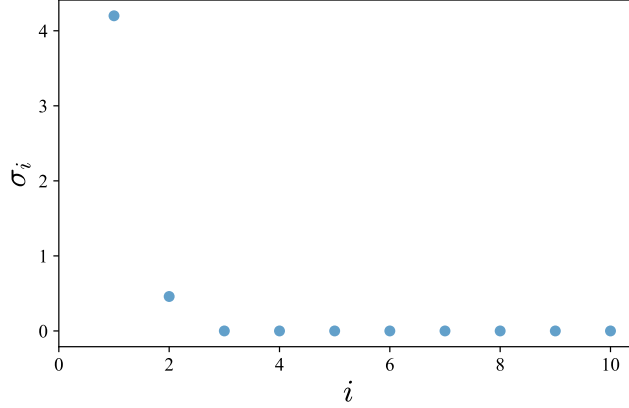


Figure 5: Scree plot for the household panel. The sharp elbow after the second singular value motivates $N = 2$.

3.5 DMD Approximation of the DSGE Model

Up to this point, we have shown that $\mathbf{B}^*$ is effectively low-rank and that rank- N DMD targets $\mathbf{B}^*$. The information projection perspective makes precise the criterion underlying this target: DMD selects the best-fitting low-rank VAR(1) projection approximation of the DSGE model’s one-step-ahead conditional dynamics. Here, two features of the DSGE model make DMD a promising prospect for an approximating model: (i) the population projection coefficient $\mathbf{B}^*$ in (35) is effectively rank-2, with household income and consumption loading predominantly on two aggregate factors $(k_{t-1}, d_{a,t})$; (ii) the infinite-order VAR coefficients $\{\mathbf{B}_j\}$ decay rapidly, making a first-order VAR a good approximation to an infinite order VAR.

We now proceed to apply DMD to a simulated household panel $\{\mathbf{y}_t\}_{t=1}^T$ and study how well it describes underlying dynamics. We begin by selecting the DMD rank N . Figure 5 presents a scree plot of singular values of the centered data matrix $\mathbf{Y}$ constructed from $\{\mathbf{y}_t\}_{t=1}^T$ by stacking the observations as columns. The sharp elbow after the second singular value suggests that $N = 2$ is a good choice. With N fixed, we first compare eigenvalues of the estimated DMD operator $\hat{\mathbf{B}}$ to those of the true

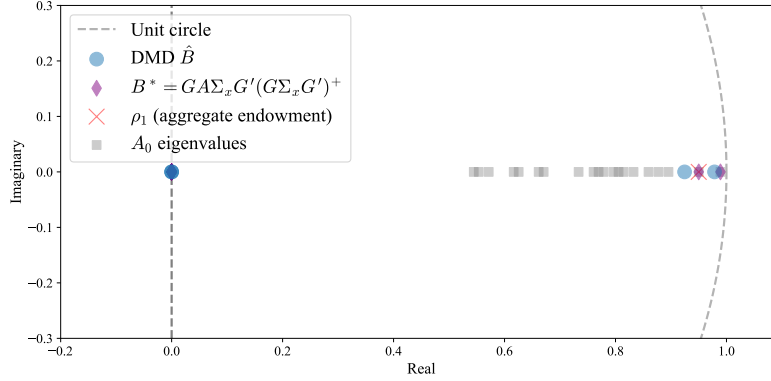


Figure 6: Recovery of eigenvalues from DMD with $N = 2$.

system. Figure 6 overlays the top 20 eigenvalues of $\mathbf{A}$, the eigenvalues of $\mathbf{B}^*$ in (35), the DMD eigenvalues, and the aggregate endowment persistence ρ_1 . With $N = 2$, the DMD eigenvalues closely match the leading eigenvalues of $\mathbf{B}^*$, confirming that DMD approximates the DSGE model's one-step linear projection dynamics well.

Recovery of eigenvalues does not by itself imply that the associated low-dimensional DMD state tracks the latent state. As we discussed in Section 3.4, we expect the leading DMD modes to track the aggregate capital and endowment components of $\mathbf{x}_t$ if DMD recovers the low-dimensional dynamics from the household panel. Figure 7 examines how well the estimated DMD mode coordinates track the aggregate state. Let $\tilde{\mathbf{x}}_t = \Phi^+ \mathbf{y}_t$ denote the DMD mode coordinates. For each mode, we compute its correlation with every component of the true state $\mathbf{x}_t$ and plot it against the state component with the highest absolute correlation (both standardized as z-scores). Indeed, the first DMD mode closely tracks aggregate capital, and the second tracks aggregate endowment with very high correlation, confirming that DMD recovers the true low-dimensional dynamics embedded in the high-dimensional panel.

Beyond latent-state recovery, we can ask whether the estimated modes reproduce the cross-sectional structure implied by the sharing rule. Figures 8 and 9 examine how DMD mode loadings vary across households. The DMD modes matrix $\Phi \in \mathbb{R}^{M \times N}$

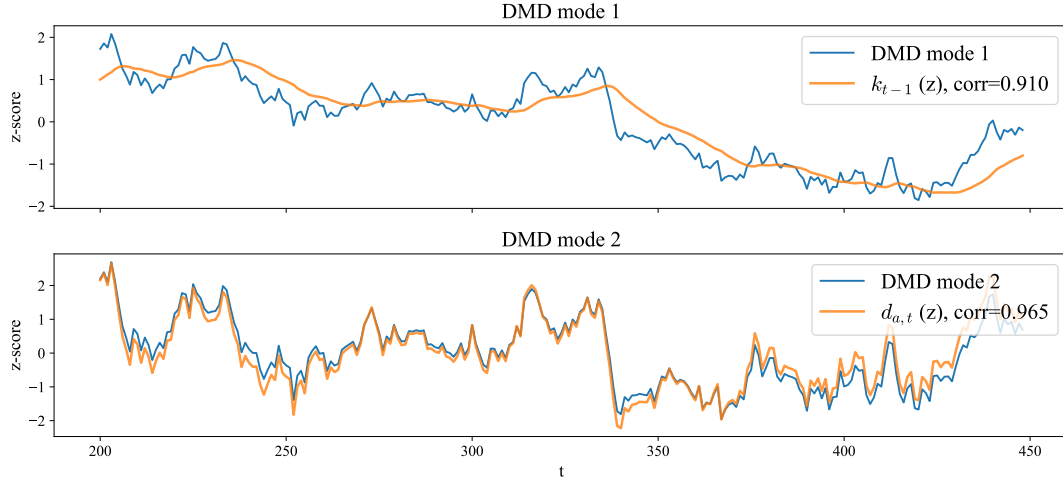


Figure 7: DMD mode time series for the stacked panel $\mathbf{y}_t$ compared with selected components of the true aggregate state $\mathbf{x}_t$.

has rows corresponding to observables. Under the sharing rule (23), the component of consumption driven by aggregate dynamics scales with μ_j , so households with larger μ_j should load more heavily on aggregate modes. Figure 8 confirms this by plotting the magnitudes of DMD loadings for pre- and post-redistribution income and for consumption against household wealth percentiles. Denote these loadings by $\Phi_{jk}^{(y_{pre})}$, $\Phi_{jk}^{(y_{post})}$, and $\Phi_{jk}^{(c)}$, respectively, where households are ordered by magnitudes of their Pareto weights μ_j . Figure 8 shows that households with higher percentiles have a larger loading on DMD modes of pre-redistribution income. Looking at post-redistribution incomes and consumptions shows how redistribution has attenuated inequality.

Figure 9 reinforces the idea that the loadings reflect the sharing rule. We extract the consumption block $\Phi^{(c)}$ and plot $|\Phi_{jk}^{(c)}|$ against household j 's Pareto weight μ_j , showing that the correlation between $|\Phi_{j1}^{(c)}|$ and μ_j is unity, indicating that the consumption-block DMD loadings are proportional to the Pareto weights $\{\mu_j\}$, as predicted by (23).

To move beyond in-sample diagnostics, Figure 10 evaluates forecasting performance

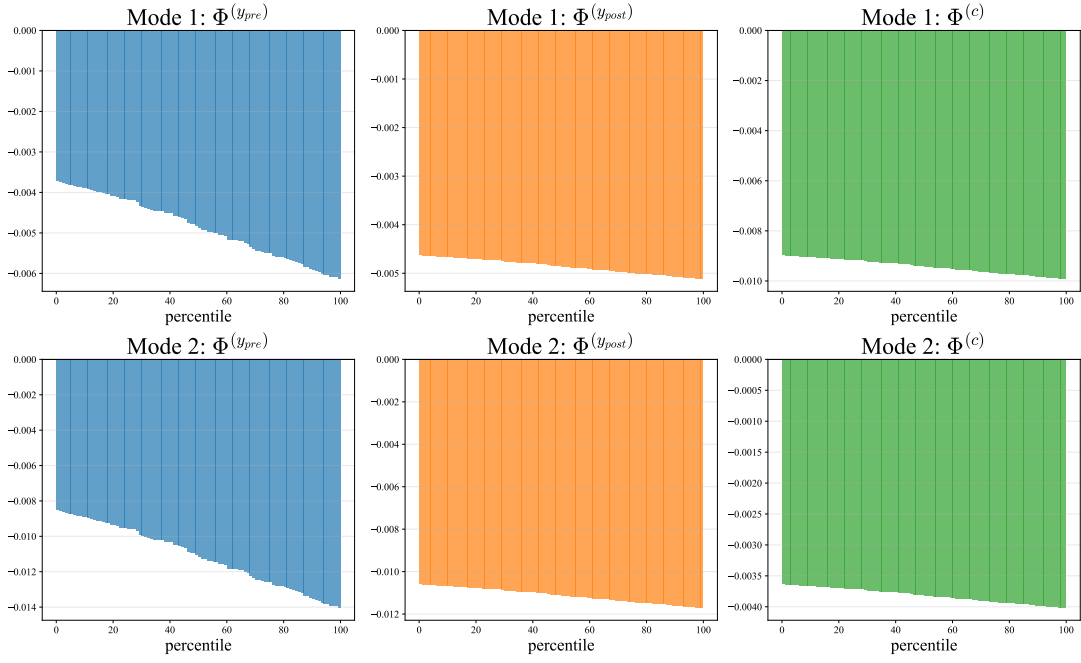


Figure 8: DMD loadings by wealth percentile for the stacked panel $\mathbf{y}_t$. Households are ordered by the Pareto weights $\{\mu_j\}$ (ascending).

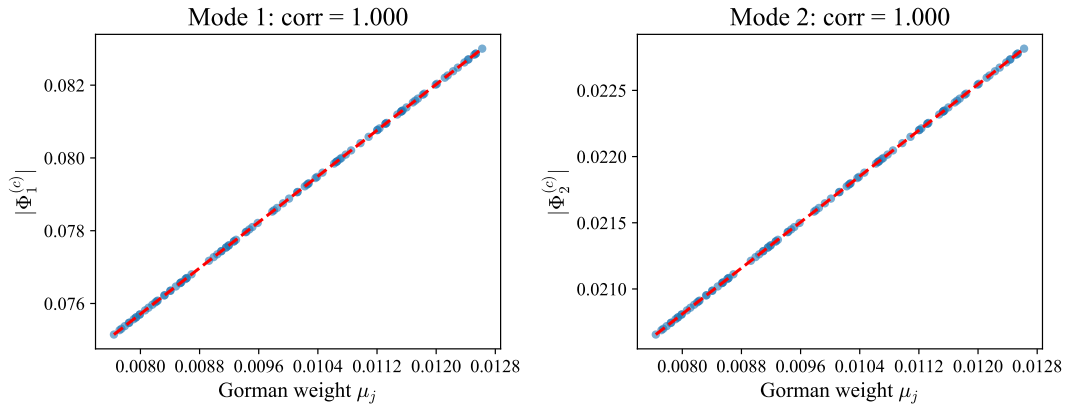


Figure 9: In the 100-household experiment, DMD loadings in the consumption block are proportional to the Pareto weights $\{\mu_j\}$, consistent with the sharing rule (23).

using a train-test split with training fraction 0.7. We estimate DMD on the training sample to obtain $\hat{\mathbf{B}}$ and the training-sample mean $\bar{\mathbf{y}}_{\text{train}}$. For each test observation $\mathbf{y}_t^{\text{GL}}$, the k -step-ahead conditional-mean forecast is

$$\hat{\mathbf{y}}_{t+k|t}^{\text{GL}} = \bar{\mathbf{y}}_{\text{train}} + \hat{\mathbf{B}}^k (\mathbf{y}_t^{\text{GL}} - \bar{\mathbf{y}}_{\text{train}}), \quad (40)$$

which iterates the estimated VAR operator k steps forward. The figure plots the cross-sectional median of both realized values $\mathbf{y}_{t+k}^{\text{GL}}$ and forecasts $\hat{\mathbf{y}}_{t+k|t}^{\text{GL}}$ for horizons $k \in \{1, 3, 5\}$. Short-horizon predictions closely track the income and consumption dynamics of the median household, while longer-horizon predictions revert toward unconditional means as the stable DMD operator is iterated forward. The same pattern holds for all cross-sectional percentiles.

Finally, our structural DSGE laboratory provides an informative numerical example that verifies key theoretical results in Section 2, particularly the implications of the rank-2 structure of $\mathbf{B}^*$ documented in Figure 4. In this rank-deficient case, $\mathbf{G}^+$ is not guaranteed to recover the full transition matrix $\mathbf{A}$ from $\mathbf{B}^*$, but a reduced two-dimensional state representation using $(k_{t-1}, d_{a,t})$ can be constructed to recover the transition matrix associated with these aggregate states. Appendix D extends these results.

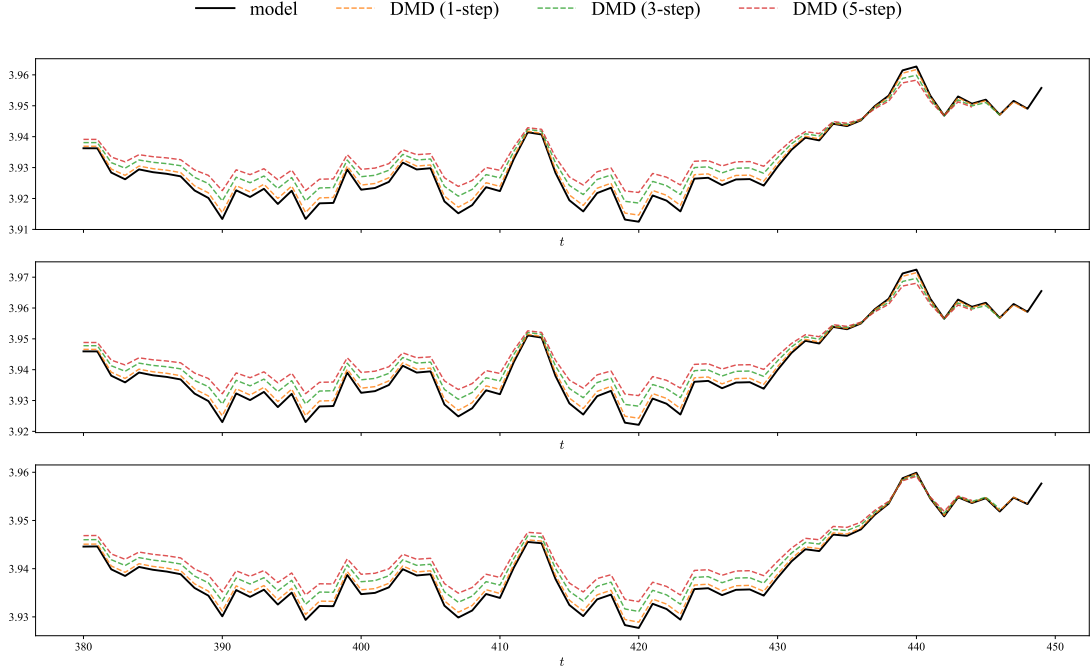


Figure 10: Out-of-sample DMD predictions in the 100-household experiment at horizons $k \in \{1, 3, 5\}$.

4 Concluding Remarks

We have studied connections between dynamic mode decompositions, reduced-rank vector autoregressions, and linear state-space models. We showed that DMD provides a computationally efficient, SVD-based estimator of low-rank first-order VAR projection coefficients in high-dimensional settings. When the measurement matrix has full column rank, the recovered nonzero eigenvalues coincide with those of the underlying state transition matrix. DMD estimates a one-step-ahead linear projection coefficient matrix. In particular, for a vector process governed by an LQG linear state-space model without observation noise, DMD is a good way to estimate a reduced-rank first-order VAR.

We study how well DMD works as an approximating model for a heterogeneous-

agent economy with complete markets and preferences satisfying Gorman aggregation conditions. From a simulated panel of household incomes and consumptions, DMD does a good job of detecting and describing low-dimensional factors that drive the cross-section. Estimated dynamic modes track the latent aggregate capital and endowment states with high correlations, and the cross-sectional loadings on these modes reveal the structural model's sharing rule.

We use information projection to study approximation properties of the DMD-based descriptive model. The estimated VAR coefficient matrix can be viewed as the Kullback–Leibler projection of the structural model's one-step-ahead conditional distribution onto a Gaussian VAR(1) family for the observables.

Our Section 3 information projection exercise illustrates how a Kepler-stage descriptive model—estimated via DMD from microeconomic panel data—can uncover economically meaningful patterns generated by a Newton-stage structural model. In our little laboratory, these patterns include a dynamic factor structure induced by homogenizing mechanisms and Pareto weights that describe the cross-section allocation in the structural DSGE model.

Appendices

A Other Applications of PCA

[Stock and Watson \(2016, Sec. 2\)](#) describe how a principal-components representation of the contemporaneous covariance matrix of a covariance-stationary vector stochastic process is an essential component of what they call a “static” representation of a “dynamic” factor model. That application of singular value decompositions and principal components analysis differs substantially from ours. To appreciate this, notice that singular value decompositions make two appearances in our calculations.¹⁹ We use a reduced SVD $\mathbf{Y} = \tilde{\mathbf{U}} \tilde{\mathbf{\Sigma}} \tilde{\mathbf{V}}^\top$ to form submatrices $\mathbf{U}$ of $\tilde{\mathbf{U}}$ and $\mathbf{V}$ of $\tilde{\mathbf{V}}$ that appear in (12). This yields the rank- N truncation $\mathbf{Y}_N := \mathbf{U} \tilde{\mathbf{\Sigma}} \mathbf{V}^\top$ that we then use to form $\hat{\mathbf{B}} = \mathbf{Y}' \mathbf{Y}_N^+$. We use $\mathbf{U}$ and $\mathbf{V}$ again when we form the reduced operator $\tilde{\mathbf{A}} = \mathbf{U}^\top \hat{\mathbf{B}} \mathbf{U} \in \mathbb{R}^{N \times N}$. At this point we compute an eigendecomposition

$$\tilde{\mathbf{A}} = \mathbf{W} \mathbf{\Lambda} \mathbf{W}^{-1},$$

where columns of the $N \times N$ matrix $\mathbf{W}$ are eigenvectors of $\tilde{\mathbf{A}}$ and eigenvalues of $\tilde{\mathbf{A}}$ appear on the diagonal of the diagonal matrix $\mathbf{\Lambda}$.

By way of contrast, [Stock and Watson](#) describe how singular value decompositions are applied to construct a “static” representation of a “dynamic” factor model by (i) assuming that an observed process $\{y_t\}$ is covariance-stationary, (ii) estimating the $M \times M$ covariance matrix C_y of y_t , (iii) using a singular value decomposition to compute M principal components of C_y , then selecting the r most important components, then (iv) adopting statistical detection procedures to infer $q < r$ “factors” in what [Stock and Watson](#) call an associated “dynamic” factor model. We do something else. In Sections 2,

¹⁹Many modern procedures compute eigendecompositions by first computing a singular value decomposition. For example, see [Strang \(2020\)](#).

the object targeted by DMD is the linear projection coefficient $\mathbf{B}^*$. The rank- N DMD operator is a low-rank estimator of this projector computed from the data matrix.

A reader of [Stock and Watson \(2016\)](#) will recognize how our procedure instead resides within a distinct tradition that connects LQG hidden Markov models and vector autoregressions to infer parameters of the hidden Markov model, a tradition in which a principal components analysis rarely makes an appearance. See especially [Stock and Watson \(2016, Sec. 2\)](#). As [Section 2.3](#) above indicates, the hidden Markov model affiliated with our DMD procedure is a very special case of the state-space models described by [Stock and Watson \(2016, Sec. 2\)](#), [Hansen and Sargent \(2013, ch. 9\)](#), and [Fernández-Villaverde et al. \(2007\)](#).

B Closed-Form Construction of $\mathbf{G}_o$

This appendix derives a closed-form expression for the measurement matrix $\mathbf{G}_o$ in [\(27\)](#).

For income, combining [\(24\)](#) with the asset decomposition $w_{j,t-1} = \omega_j k_{t-1} + \hat{k}_{j,t-1}$ gives

$$y_{jt}(\omega) = \omega_j d_t + (R-1)\omega_j k_{t-1} + (R-1)\hat{k}_{j,t-1}, \quad (41)$$

where the first two terms depend only on aggregates, but the third term involves household j 's idiosyncratic bond position $\hat{k}_{j,t-1}$. In the limited-markets implementation of the complete-markets allocation, the bond position is driven by the sharing-rule deviation through the law of motion

$$\hat{k}_{jt} = R\hat{k}_{j,t-1} - \tilde{\chi}_{jt}, \quad (42)$$

where $\tilde{\chi}_{jt}$ is the deviation term from the sharing rule [\(23\)](#). The initial bond position is

pinned down by the present value of future deviations:

$$\hat{k}_{j0} = \sum_{s=1}^{\infty} \beta^s \tilde{\chi}_{js}. \quad (43)$$

In general, writing y_{jt} as a static function of $\mathbf{x}_t$ alone requires augmenting the state with $\{\hat{k}_{j,t-1}\}_{j=1}^J$. However, when idiosyncratic preference shocks are muted so that $\tilde{\chi}_{jt} = 0$, equations (42)–(43) imply $\hat{k}_{jt} = 0$ for all j and t , and income can be written as

$$y_{jt}(\boldsymbol{\omega}) = \omega_j \mathbf{S}_d \mathbf{x}_t + (R - 1)\omega_j \mathbf{S}_k \mathbf{x}_t, \quad (44)$$

where $\mathbf{S}_d$ and $\mathbf{S}_k$ are selection vectors such that $d_t = \mathbf{S}_d \mathbf{x}_t$ and $k_{t-1} = \mathbf{S}_k \mathbf{x}_t$. This yields the income row $\mathbf{G}_j^y = \omega_j \mathbf{S}_d + (R - 1)\omega_j \mathbf{S}_k$ with $\mathbf{G}^y \in \{\mathbf{G}^{y,\text{pre}}, \mathbf{G}^{y,\text{post}}\}$ and $\omega_j \in \{\mu_j, \mu_j^*\}$ correspondingly.

With $\tilde{\chi}_{jt} = 0$, the sharing rule (23) reduces to $c_{jt} = \mu_j c_t$. Since $c_t = \mathbf{S}_c \mathbf{x}_t$ for a selection vector $\mathbf{S}_c$, this yields the consumption row $\mathbf{G}_j^c = \mu_j \mathbf{S}_c$ and hence $\mathbf{G}_j^{c,\text{post}} = \mu_j^* \mathbf{S}_c$ after redistribution.

C Additional Results

The results below supplement the body of the paper. We first record identities for the projection coefficient $\mathbf{B}^*$ when the measurement map may be rank-deficient and useful identities associated with the Kalman gain in full column rank regime. We then provide complementary state-space representations for DMD output that relax the assumptions in Section 2.2. Proofs are collected in Appendix C.5.

C.1 Rank-Deficient Measurement

The first result records the form of the linear projection coefficient $\mathbf{B}^*$ when the measurement map is not necessarily injective.

Proposition 3. Let $\Sigma_x := \text{Cov}(\mathbf{x}_t)$ under (1) and let $\mathbf{B}^*$ be defined by (7). Define $\mathbf{G}_{\Sigma_x}^+ := \Sigma_x \mathbf{G}^\top (\mathbf{G} \Sigma_x \mathbf{G}^\top)^+$. Then

(i) $\mathbf{B}^*$ satisfies

$$\mathbf{B}^* = \mathbf{G} \mathbf{A} \Sigma_x \mathbf{G}^\top (\mathbf{G} \Sigma_x \mathbf{G}^\top)^+ = \mathbf{G} \mathbf{A} \mathbf{G}_{\Sigma_x}^+,$$

(ii) The following properties hold:

$$\text{range}(\mathbf{B}^*) \subseteq \text{range}(\mathbf{G}), \quad \text{rank}(\mathbf{B}^*) \leq \text{rank}(\mathbf{G}),$$

$$\mathbf{B}^* \mathbf{v} = \lambda \mathbf{v}, \lambda \neq 0 \Rightarrow \mathbf{v} \in \text{range}(\mathbf{G}).$$

Proof. See Appendix C.5.3. □

Remark 1. Proposition 3 establishes that $\mathbf{B}^* = \mathbf{G} \mathbf{A} \mathbf{G}_{\Sigma_x}^+$ regardless of $\text{rank}(\mathbf{G})$, where $\mathbf{B}^*$ is the population one-step linear projection coefficient in (7); see also equation (35). When $\text{rank}(\mathbf{G}) < N_x$, the full N_x -dimensional transition matrix $\mathbf{A}$ cannot be recovered from the observables $\{\mathbf{y}_t\}$, but a rank- $\text{rank}(\mathbf{G})$ DMD suffices to recover the dominant eigenvalues and modes of the panel that are observable through $\mathbf{G}$.

C.2 Additional Identities under Full-Column-Rank Measurement

The second result records an identity for the steady-state covariance and Kalman gain in the innovations representation under full-column-rank measurement.

Corollary 1. Suppose that $\mathbf{G}$ has full column rank. Then the Riccati equation (5) reduces to

$$\Sigma_\infty = \mathbf{C} \mathbf{C}^\top, \tag{45}$$

If in addition $\mathbf{C} \mathbf{C}^\top \succ 0$, then $\Sigma_\infty \succ 0$, and the corresponding Kalman gain satisfies $\mathbf{A} = \mathbf{K} \mathbf{G}$.

Proof. See Appendix C.5.4. □

Under the assumptions of Corollary 1 and Proposition 1(ii), we also obtain $\mathbf{B}^* = \mathbf{G} \mathbf{K}$. Indeed, when $\Sigma_\infty \succ 0$ and $\text{rank}(\mathbf{G}) = N_x$,

$$(\mathbf{G} \Sigma_\infty \mathbf{G}^\top)^+ = (\mathbf{G}^+)^{\top} \Sigma_\infty^{-1} \mathbf{G}^+,$$

so

$$\mathbf{G} \mathbf{K} = \mathbf{G} \mathbf{A} \Sigma_\infty \mathbf{G}^\top (\mathbf{G} \Sigma_\infty \mathbf{G}^\top)^+ = \mathbf{G} \mathbf{A} \mathbf{G}^+ = \mathbf{B}^*,$$

where the last equality uses the same derivation leading to Proposition 1(ii).

C.3 State-Space Representation Induced by $\hat{\mathbf{B}}$

This result provides a reduced-coordinate representation for the fitted reduced-rank VAR associated with the least-squares estimator $\hat{\mathbf{B}}$, and records the orthogonality result of the fitted residuals.

Proposition 4. *Define*

$$\hat{\mathbf{B}} := \mathbf{Y}' \mathbf{V} \Sigma^{-1} \mathbf{U}^\top, \quad \tilde{\mathbf{A}} := \mathbf{U}^\top \mathbf{Y}' \mathbf{V} \Sigma^{-1}.$$

Assume $\tilde{\mathbf{A}}$ is diagonalizable with $\tilde{\mathbf{A}} = \mathbf{W} \mathbf{\Lambda} \mathbf{W}^{-1}$, and define DMD modes $\Phi := \mathbf{Y}' \mathbf{V} \Sigma^{-1} \mathbf{W}$. For $t = 2, \dots, T+1$, define residuals

$$\hat{\mathbf{a}}_t := \mathbf{y}_t - \hat{\mathbf{B}} \mathbf{y}_{t-1}, \quad \hat{\Omega} := \frac{1}{T} \sum_{t=2}^{T+1} \hat{\mathbf{a}}_t \hat{\mathbf{a}}_t^\top.$$

Define reduced coordinates

$$\mathbf{x}_t^{\text{red}} := \mathbf{W}^{-1} \mathbf{U}^\top \mathbf{y}_t.$$

Then for $t = 1, \dots, T$, the fitted reduced-rank VAR admits the state-space representation

$$\begin{aligned}\mathbf{x}_{t+1}^{\text{red}} &= \mathbf{\Lambda} \mathbf{x}_t^{\text{red}} + \mathbf{W}^{-1} \mathbf{U}^\top \hat{\mathbf{a}}_{t+1}, \\ \mathbf{y}_{t+1} &= \mathbf{\Phi} \mathbf{x}_t^{\text{red}} + \hat{\mathbf{a}}_{t+1},\end{aligned}\tag{46}$$

where $\sum_{t=1}^T \hat{\mathbf{a}}_{t+1} (\mathbf{U}^\top \mathbf{y}_t)^\top = \mathbf{0}$.

Proof. See Appendix C.5.5. □

It is important to note that $\hat{\mathbf{a}}_t$ are defined by the least-squares projection used to fit the VAR, so they impose only the sample orthogonality conditions $\sum_{t=2}^{T+1} \hat{\mathbf{a}}_t (\mathbf{U}^\top \mathbf{y}_{t-1})^\top = \mathbf{0}$ (and, in population, $\mathbb{E}[\hat{\mathbf{a}}_t (\mathbf{U}^\top \mathbf{y}_{t-1})^\top] = \mathbf{0}$), whereas the innovations $\mathbf{a}_t$ in (3) satisfy $\mathbf{a}_t \perp \mathbf{y}^{t-1}$. When the data-generating process is rank- N VAR(1) (for example, under the full-column-rank measurement condition in Proposition 1(i)), these notions coincide and (17) becomes an innovations representation: the fitted reduced-rank VAR only uses the N -dimensional lag subspace spanned by the left singular vectors $\mathbf{U}$, so the least-squares residuals $\hat{\mathbf{a}}_t$ are orthogonal only to the compressed regressors $\mathbf{U}^\top \mathbf{y}_{t-1}$, but under a rank- N VAR(1) the lagged observations $\mathbf{y}_{t-1}$ lie in $\text{col}(\mathbf{U})$, hence $\mathbf{U}^\top$ is a lossless coordinate change on that subspace and $\hat{\mathbf{a}}_t \perp \mathbf{U}^\top \mathbf{y}_{t-1}$ is equivalent to $\hat{\mathbf{a}}_t \perp \mathbf{y}_{t-1}$, so $\hat{\mathbf{a}}_t$ matches the innovations $\mathbf{a}_t$. Similar argument applies to the mode-space representation in Appendix C.4 below.

C.4 Mode-Space Representation

This result gives a mode-space representation for $\tilde{\mathbf{B}}$ that remains valid even when the DMD modes matrix $\mathbf{\Phi}$ is not full column rank.

Corollary 2. *Maintain the setup of Proposition 4 and define*

$$\tilde{\mathbf{B}} := \mathbf{\Phi} \mathbf{\Lambda} \mathbf{\Phi}^+.$$

For $t = 2, \dots, T + 1$, define the $\tilde{\mathbf{B}}$ -residuals

$$\tilde{\mathbf{a}}_t := \mathbf{y}_t - \tilde{\mathbf{B}} \mathbf{y}_{t-1}, \quad \tilde{\mathbf{\Omega}} := \frac{1}{T} \sum_{t=2}^{T+1} \tilde{\mathbf{a}}_t \tilde{\mathbf{a}}_t^\top,$$

and define mode coordinates $\tilde{\mathbf{x}}_t := \mathbf{\Phi}^+ \mathbf{y}_t$. Then for $t = 1, \dots, T$, the reduced-rank VAR driven by $\tilde{\mathbf{B}}$ admits the linear state-space representation

$$\begin{aligned} \tilde{\mathbf{x}}_{t+1} &= (\mathbf{\Phi}^+ \mathbf{\Phi}) \mathbf{\Lambda} \tilde{\mathbf{x}}_t + \mathbf{\Phi}^+ \tilde{\mathbf{a}}_{t+1}, \\ \mathbf{y}_{t+1} &= \mathbf{\Phi} \mathbf{\Lambda} \tilde{\mathbf{x}}_t + \tilde{\mathbf{a}}_{t+1}. \end{aligned} \tag{47}$$

If in addition, $\mathbf{\Phi}$ has full column rank, then (15) holds.

Proof. See Appendix C.5.6. □

C.5 Proofs

C.5.1 Proposition 1

Proof. Regarding (i), since $\mathbf{G}$ has full column rank, $\mathbf{G}^+ \mathbf{G} = \mathbf{I}_{N_x}$ and therefore $\mathbf{x}_{t-1} = \mathbf{G}^+ \mathbf{y}_{t-1}$. Using $\mathbf{x}_t = \mathbf{A} \mathbf{x}_{t-1} + \mathbf{C} \mathbf{w}_t$ gives

$$\mathbf{y}_t = \mathbf{G} \mathbf{x}_t = \mathbf{G}(\mathbf{A} \mathbf{x}_{t-1} + \mathbf{C} \mathbf{w}_t) = \mathbf{G} \mathbf{A} \mathbf{G}^+ \mathbf{y}_{t-1} + \mathbf{G} \mathbf{C} \mathbf{w}_t = \bar{\mathbf{B}} \mathbf{y}_{t-1} + \mathbf{u}_t.$$

Because $\{\mathbf{w}_t\}$ is i.i.d. and independent of the past $\mathbf{y}^{t-1}$, the same holds for $\{\mathbf{u}_t\}$, so $\mathbf{u}_t \perp \mathbf{y}^{t-1}$. Hence $\mathbb{E}[\mathbf{y}_t | \mathbf{y}^{t-1}] = \bar{\mathbf{B}} \mathbf{y}_{t-1}$ and $\mathbf{a}_t = \mathbf{y}_t - \mathbb{E}[\mathbf{y}_t | \mathbf{y}^{t-1}] = \mathbf{u}_t$. This completes the first part of (i).

Let $\mathbf{\Gamma}_0 := \text{Cov}(\mathbf{y}_{t-1})$ and $\mathbf{\Gamma}_1 := \text{Cov}(\mathbf{y}_t, \mathbf{y}_{t-1})$. Orthogonality implies $\bar{\mathbf{B}} \mathbf{\Gamma}_0 = \mathbf{\Gamma}_1$.²⁰ By definition $\mathbf{B}^* = \mathbf{\Gamma}_1 \mathbf{\Gamma}_0^+$, so $(\mathbf{B}^* - \bar{\mathbf{B}}) \mathbf{\Gamma}_0 = 0$. Since $\mathbf{y}_{t-1} \in \text{range}(\mathbf{\Gamma}_0)$ a.s., we have $(\mathbf{B}^* - \bar{\mathbf{B}}) \mathbf{y}_{t-1} = 0$ a.s. and hence statement (i) follows.

²⁰This can be seen from the fact that $\mathbf{\Gamma}_1 = \mathbb{E}[\mathbf{y}_t \mathbf{y}_{t-1}^\top] = \mathbb{E}[(\bar{\mathbf{B}} \mathbf{y}_{t-1} + \mathbf{u}_t) \mathbf{y}_{t-1}^\top] = \bar{\mathbf{B}} \mathbb{E}[\mathbf{y}_{t-1} \mathbf{y}_{t-1}^\top] + \mathbb{E}[\mathbf{u}_t \mathbf{y}_{t-1}^\top] = \bar{\mathbf{B}} \mathbf{\Gamma}_0 + 0$.

For (ii), by Proposition 3(i), $\mathbf{B}^* = \mathbf{G} \mathbf{A} \Sigma_x \mathbf{G}^\top (\mathbf{G} \Sigma_x \mathbf{G}^\top)^+$. If $\Sigma_x \succ 0$ and $\mathbf{G}$ has full column rank, then

$$(\mathbf{G} \Sigma_x \mathbf{G}^\top)^+ = (\mathbf{G}^+)^{\top} \Sigma_x^{-1} \mathbf{G}^+,$$

so $\mathbf{B}^* = \mathbf{G} \mathbf{A} \Sigma_x \mathbf{G}^\top (\mathbf{G}^+)^{\top} \Sigma_x^{-1} \mathbf{G}^+ = \mathbf{G} \mathbf{A} \mathbf{G}^+ = \bar{\mathbf{B}}$.²¹ Then $\mathbf{A} = \mathbf{G}^+ \mathbf{B}^* \mathbf{G}$ follows from $\mathbf{G}^+ \mathbf{G} = \mathbf{I}_{N_x}$. $\square$

C.5.2 Proposition 2

Proof. Recall that $\mathbf{Y}_N := \mathbf{U} \Sigma \mathbf{V}^\top$ denotes the rank- N truncated SVD of the data matrix and $\sigma_N > 0$. Then

$$\mathbf{Y}_N^+ = \mathbf{V} \Sigma^{-1} \mathbf{U}^\top, \quad \hat{\mathbf{B}} = \mathbf{Y}' \mathbf{Y}_N^+ = \mathbf{Y}' \mathbf{V} \Sigma^{-1} \mathbf{U}^\top.$$

Define the reduced operator and the DMD modes by

$$\tilde{\mathbf{A}} := \mathbf{U}^\top \mathbf{Y}' \mathbf{V} \Sigma^{-1}, \quad \tilde{\mathbf{A}} \mathbf{W} = \mathbf{W} \Lambda, \quad \Phi := \mathbf{Y}' \mathbf{V} \Sigma^{-1} \mathbf{W}.$$

Then

$$\hat{\mathbf{B}} \Phi = \mathbf{Y}' \mathbf{V} \Sigma^{-1} (\mathbf{U}^\top \mathbf{Y}' \mathbf{V} \Sigma^{-1}) \mathbf{W} = \mathbf{Y}' \mathbf{V} \Sigma^{-1} (\tilde{\mathbf{A}} \mathbf{W}) = \mathbf{Y}' \mathbf{V} \Sigma^{-1} \mathbf{W} \Lambda = \Phi \Lambda.$$

This proves statement (i).

When restricting $\hat{\mathbf{B}}$ to act on $\text{col}(\Phi)$, equivalently replace $\hat{\mathbf{B}}$ by $\hat{\mathbf{B}} \mathbf{P}_\Phi$. Using $\mathbf{Y}_N^+ \Phi = \mathbf{V} \Sigma^{-1} \mathbf{U}^\top \Phi = \mathbf{V} \Sigma^{-1} \tilde{\mathbf{A}} \mathbf{W} = \mathbf{V} \Sigma^{-1} \mathbf{W} \Lambda$, we have

$$\hat{\mathbf{B}} \mathbf{P}_\Phi = \Phi \Lambda \Phi^+,$$

and hence $\hat{\mathbf{B}}$ and $\Phi \Lambda \Phi^+$ coincide on $\text{col}(\Phi)$.²² This completes the proof. $\square$

²¹Because $\mathbf{G}^\top (\mathbf{G}^+)^{\top} = (\mathbf{G}^+ \mathbf{G})^\top = \mathbf{I}_{N_x}$.

²²Moreover, among all $\mathbf{B}$ with $\mathbf{B} \Phi = \Phi \Lambda$, the unique minimizer of the Frobenius norm $\|\mathbf{B}\|_F$ is $\Phi \Lambda \Phi^+$. This

C.5.3 Proposition 3

Proof. For (i), since $\mathbf{y}_t = \mathbf{G} \mathbf{x}_t$ and $\mathbf{x}_t$ is stationary,

$$\text{Cov}(\mathbf{y}_t, \mathbf{y}_{t-1}) = \mathbf{G} \mathbf{A} \boldsymbol{\Sigma}_x \mathbf{G}^\top, \quad \text{Cov}(\mathbf{y}_{t-1}) = \mathbf{G} \boldsymbol{\Sigma}_x \mathbf{G}^\top.$$

Substituting into (7) yields $\mathbf{B}^* = \mathbf{G} \mathbf{A} \boldsymbol{\Sigma}_x \mathbf{G}^\top (\mathbf{G} \boldsymbol{\Sigma}_x \mathbf{G}^\top)^+ = \mathbf{G} \mathbf{A} \mathbf{G}_{\boldsymbol{\Sigma}_x}^+$.

(ii) follows from $\mathbf{B}^* = \mathbf{G}(\mathbf{A} \mathbf{G}_{\boldsymbol{\Sigma}_x}^+)$, and if $\mathbf{B}^* \mathbf{v} = \lambda \mathbf{v}$ with $\lambda \neq 0$ then $\mathbf{v} = (1/\lambda) \mathbf{B}^* \mathbf{v} \in \text{range}(\mathbf{G})$. $\square$

C.5.4 Corollary 1

Proof. Fix any symmetric $\boldsymbol{\Sigma} \succeq 0$ and let $\boldsymbol{\Sigma}^{1/2}$ denote its symmetric square root. Define $\mathbf{Q} := \mathbf{G} \boldsymbol{\Sigma}^{1/2}$. Since $\mathbf{G}$ has full column rank, $\mathbf{G}^\top$ is surjective, and therefore

$$\text{range}(\mathbf{Q}^\top) = \text{range}(\boldsymbol{\Sigma}^{1/2} \mathbf{G}^\top) = \text{range}(\boldsymbol{\Sigma}^{1/2}).$$

Hence $\mathbf{Q}^\top (\mathbf{Q} \mathbf{Q}^\top)^+ \mathbf{Q}$ is the orthogonal projector onto $\text{range}(\boldsymbol{\Sigma}^{1/2})$, so it acts as the identity on $\text{range}(\boldsymbol{\Sigma}^{1/2})$. Hence,

$$\boldsymbol{\Sigma} = \boldsymbol{\Sigma}^{1/2} \mathbf{Q}^\top (\mathbf{Q} \mathbf{Q}^\top)^+ \mathbf{Q} \boldsymbol{\Sigma}^{1/2} = \boldsymbol{\Sigma} \mathbf{G}^\top (\mathbf{G} \boldsymbol{\Sigma} \mathbf{G}^\top)^+ \mathbf{G} \boldsymbol{\Sigma}.$$

Applying this identity with $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}_\infty$ in (5) yields

$$\boldsymbol{\Sigma}_\infty = \mathbf{A} \boldsymbol{\Sigma}_\infty \mathbf{A}^\top + \mathbf{C} \mathbf{C}^\top - \mathbf{A} \boldsymbol{\Sigma}_\infty \mathbf{G}^\top (\mathbf{G} \boldsymbol{\Sigma}_\infty \mathbf{G}^\top)^+ \mathbf{G} \boldsymbol{\Sigma}_\infty \mathbf{A}^\top = \mathbf{C} \mathbf{C}^\top,$$

which proves (45). Under the additional assumption $\mathbf{C} \mathbf{C}^\top \succ 0$, we have $\boldsymbol{\Sigma}_\infty \succ 0$.

Letting $\mathbf{Q} := \mathbf{G} \boldsymbol{\Sigma}_\infty^{1/2}$, $\mathbf{Q}$ has full column rank, so $\mathbf{Q}^\top (\mathbf{Q} \mathbf{Q}^\top)^+ \mathbf{Q} = \mathbf{I}_{N_x}$, which is

follows from the KKT conditions for $\min_{\mathbf{B}} \frac{1}{2} \|\mathbf{B}\|_F^2$ subject to $\mathbf{B} \boldsymbol{\Phi} = \boldsymbol{\Phi} \boldsymbol{\Lambda}$.

equivalent to

$$\Sigma_\infty \mathbf{G}^\top (\mathbf{G} \Sigma_\infty \mathbf{G}^\top)^+ \mathbf{G} = \mathbf{I}_{N_x}.$$

Multiplying (4) on the right by $\mathbf{G}$ gives $\mathbf{K} \mathbf{G} = \mathbf{A}$. □

C.5.5 Proposition 4

Proof. We prove the measurement and state equations first. By definition, $\mathbf{y}_{t+1} = \hat{\mathbf{B}} \mathbf{y}_t + \hat{\mathbf{a}}_{t+1}$. Using $\Phi = \mathbf{Y}' \mathbf{V} \Sigma^{-1} \mathbf{W}$ and $\hat{\mathbf{B}} = \mathbf{Y}' \mathbf{V} \Sigma^{-1} \mathbf{U}^\top$, we have $\Phi \mathbf{x}_t^{\text{red}} = \mathbf{Y}' \mathbf{V} \Sigma^{-1} \mathbf{W} (\mathbf{W}^{-1} \mathbf{U}^\top \mathbf{y}_t) = \hat{\mathbf{B}} \mathbf{y}_t$, which gives the measurement equation $\mathbf{y}_{t+1} = \Phi \mathbf{x}_t^{\text{red}} + \hat{\mathbf{a}}_{t+1}$.

Premultiplying $\mathbf{y}_{t+1} = \Phi \mathbf{x}_t^{\text{red}} + \hat{\mathbf{a}}_{t+1}$ by $\mathbf{U}^\top$ yields

$$\begin{aligned} \mathbf{U}^\top \mathbf{y}_{t+1} &= \mathbf{U}^\top \Phi \mathbf{x}_t^{\text{red}} + \mathbf{U}^\top \hat{\mathbf{a}}_{t+1} \\ &= (\mathbf{U}^\top \mathbf{Y}' \mathbf{V} \Sigma^{-1} \mathbf{W}) \mathbf{x}_t^{\text{red}} + \mathbf{U}^\top \hat{\mathbf{a}}_{t+1} \\ &= \tilde{\mathbf{A}} \mathbf{W} \mathbf{x}_t^{\text{red}} + \mathbf{U}^\top \hat{\mathbf{a}}_{t+1} \\ &= \mathbf{W} \Lambda \mathbf{x}_t^{\text{red}} + \mathbf{U}^\top \hat{\mathbf{a}}_{t+1}. \end{aligned}$$

Multiplying by $\mathbf{W}^{-1}$ gives the state equation $\mathbf{x}_{t+1}^{\text{red}} = \Lambda \mathbf{x}_t^{\text{red}} + \mathbf{W}^{-1} \mathbf{U}^\top \hat{\mathbf{a}}_{t+1}$.

For the orthogonality, let $\hat{\mathbf{A}} := [\hat{\mathbf{a}}_2, \dots, \hat{\mathbf{a}}_{T+1}] = \mathbf{Y}' - \hat{\mathbf{B}} \mathbf{Y}$. Using $\mathbf{U}^\top \mathbf{Y} = \Sigma \mathbf{V}^\top$,

$$\hat{\mathbf{B}} \mathbf{Y} = \mathbf{Y}' \mathbf{V} \Sigma^{-1} (\mathbf{U}^\top \mathbf{Y}) = \mathbf{Y}' \mathbf{V} \Sigma^{-1} (\Sigma \mathbf{V}^\top) = \mathbf{Y}' \mathbf{V} \mathbf{V}^\top,$$

so $\hat{\mathbf{A}} = \mathbf{Y}' (\mathbf{I}_T - \mathbf{V} \mathbf{V}^\top)$. Also $(\mathbf{U}^\top \mathbf{Y})^\top = (\Sigma \mathbf{V}^\top)^\top = \mathbf{V} \Sigma$, hence

$$\hat{\mathbf{A}} (\mathbf{U}^\top \mathbf{Y})^\top = \mathbf{Y}' (\mathbf{I}_T - \mathbf{V} \mathbf{V}^\top) \mathbf{V} \Sigma = \mathbf{Y}' (\mathbf{V} - \mathbf{V}) \Sigma = \mathbf{0},$$

which is equivalent to $\sum_{t=1}^T \hat{\mathbf{a}}_{t+1} (\mathbf{U}^\top \mathbf{y}_t)^\top = \mathbf{0}$. □

C.5.6 Corollary 2

Proof. By definition of $\tilde{\mathbf{a}}_{t+1}$, we have

$$\mathbf{y}_{t+1} = \tilde{\mathbf{B}} \mathbf{y}_t + \tilde{\mathbf{a}}_{t+1} = \Phi \Lambda \Phi^+ \mathbf{y}_t + \tilde{\mathbf{a}}_{t+1} = \Phi \Lambda \tilde{\mathbf{x}}_t + \tilde{\mathbf{a}}_{t+1},$$

which gives the measurement equation. Premultiplying by Φ^+ yields

$$\tilde{\mathbf{x}}_{t+1} = \Phi^+ \Phi \Lambda \tilde{\mathbf{x}}_t + \Phi^+ \tilde{\mathbf{a}}_{t+1},$$

which gives (47). When Φ has full column rank, $\Phi^+ \Phi = \mathbf{I}_N$, yielding (15). $\square$

D Two-State Representation in DSGE Application

In the full system with the high-dimensional aggregate state, the centered state has dimension $N_x = 54$ and the measurement matrix has numerical rank $\text{rank}(\mathbf{G}) = 2 < N_x$, consistent with the fact that the observable dynamics are effectively two-dimensional as discussed in Section 3.3 (see Figure 4). In this appendix, we focus on verifying the theoretical implications of rank-deficient measurement in this DSGE application. Specifically, we compute objects implied by the theory by replacing the full state representation in (1) with a reduced two-state representation that only tracks the aggregate capital and aggregate endowment states.

Recall that with $\Sigma_x = \text{Cov}(\mathbf{x}_t)$, the projection coefficient in (35) can be written as

$$\mathbf{B}^* = \mathbf{G} \mathbf{A} \mathbf{G}_{\Sigma_x}^+, \quad \text{where} \quad \mathbf{G}_{\Sigma_x}^+ := \Sigma_x \mathbf{G}^\top (\mathbf{G} \Sigma_x \mathbf{G}^\top)^+ \quad (48)$$

is the Σ_x -weighted pseudoinverse of $\mathbf{G}$. Equation (48) is simply a re-expression of Proposition 3(i).

When $\text{rank}(\mathbf{G}) = N_x$ and $\Sigma_x \succ 0$, Proposition 1(ii) implies that $\mathbf{G}_{\Sigma_x}^+ = \mathbf{G}^+$ and

hence $\mathbf{B}^* = \mathbf{G} \mathbf{A} \mathbf{G}^+$. When $\text{rank}(\mathbf{G}) < N_x$, (1) is partially observed and $\mathbf{G}_{\Sigma_x}^+$ need not equal $\mathbf{G}^+$. In general, the simplification $\mathbf{B}^* = \mathbf{G} \mathbf{A} \mathbf{G}^+$ fails and $\mathbf{A}$ cannot be recovered from $(\mathbf{B}^*, \mathbf{G})$ via $\mathbf{A} = \mathbf{G}^+ \mathbf{B}^* \mathbf{G}$.

We verify these theoretical results numerically in the Gorman experiment. Table 1 compares $\mathbf{B}^*$ to the projections $\mathbf{G} \mathbf{A} \mathbf{G}^+$ and $\mathbf{G} \mathbf{A} \mathbf{G}_{\Sigma_x}^+$. Consistent with Proposition 3, $\mathbf{B}^*$ is matched by $\mathbf{G} \mathbf{A} \mathbf{G}_{\Sigma_x}^+$ up to small relative error, while $\mathbf{G} \mathbf{A} \mathbf{G}^+$ can deviate sharply when $\mathbf{G}$ is rank-deficient.

Table 1: Comparison of population equalities and estimates in the Gorman experiment

	Minimal state ($k_{t-1}, d_{a,t}$)	Full state (54-dimensional)
State dimension N_x	2	54
Numerical rank($\mathbf{G}$)	2 (full)	2 (deficient)
$\ \mathbf{B}^* - \mathbf{G} \mathbf{A} \mathbf{G}^+\ _F / \ \mathbf{B}^*\ _F$	1.9×10^{-14}	10.1
$\ \mathbf{B}^* - \mathbf{G} \mathbf{A} \mathbf{G}_{\Sigma_x}^+\ _F / \ \mathbf{B}^*\ _F$	1.9×10^{-14}	1.9×10^{-14}
$\ \mathbf{G}^+ \mathbf{B}^* \mathbf{G} - \mathbf{A}\ _F / \ \mathbf{A}\ _F$	8.9×10^{-15}	0.899
$\ \mathbf{G}^+ \hat{\mathbf{B}} \mathbf{G} - \mathbf{A}\ _F / \ \mathbf{A}\ _F$	0.0254	0.937

Note: In the minimal-state column, $(\mathbf{A}, \mathbf{G})$ refers to the reduced matrices $(\mathbf{A}_{\min}, \mathbf{G}_{\min})$, while the full-state column uses the matrices $(\mathbf{A}, \mathbf{G})$ from Section 3.4.

To illustrate the full-column-rank case, we also construct a reduced (“minimal”) state representation using only the two aggregate factors that drive the centered panel, namely, aggregate capital k_{t-1} and the aggregate endowment state $d_{a,t}$. Let $\mathbf{x}_t^{\min} := (k_{t-1}, d_{a,t})^\top$ and write the centered panel as $\mathbf{y}_t = \mathbf{G}_{\min} \mathbf{x}_t^{\min}$, where $\mathbf{G}_{\min} \in \mathbb{R}^{M \times 2}$ has full column rank. In our experiment, $(M, T) = (300, 250)$, so $\mathbf{G}_{\min}$ has shape 300×2 and the full data matrix $[\mathbf{y}_1, \dots, \mathbf{y}_{T+1}]$ has shape 300×251 . Applying DMD to $\{\mathbf{y}_t\}$ in this minimal-state representation yields $\|\mathbf{B}^* - \mathbf{G}_{\min} \mathbf{A}_{\min} \mathbf{G}_{\min}^+\|_F / \|\mathbf{B}^*\|_F = 1.9 \times 10^{-14}$ and $\|\hat{\mathbf{B}} - \mathbf{B}^*\|_F / \|\mathbf{B}^*\|_F = 2.94 \times 10^{-2}$. The recovered transition matrix is accurate up to machine precision with $\|\mathbf{G}_{\min}^+ \mathbf{B}^* \mathbf{G}_{\min} - \mathbf{A}_{\min}\|_F / \|\mathbf{A}_{\min}\|_F = 8.9 \times 10^{-15}$. By contrast, a naive attempt to recover the full high-dimensional transition matrix via $\mathbf{A} = \mathbf{G}^+ \mathbf{B}^* \mathbf{G}$ produces large errors in this experiment $\|\mathbf{G}^+ \mathbf{B}^* \mathbf{G} - \mathbf{A}\|_F / \|\mathbf{A}\|_F \approx 0.899$. Similarly,

using the DMD estimator $\hat{\mathbf{B}}$ instead of $\mathbf{B}^*$, the minimal transition matrix estimate is given by

$$\hat{\mathbf{A}}_{\min} := \mathbf{G}_{\min}^+ \hat{\mathbf{B}} \mathbf{G}_{\min} = \begin{bmatrix} 0.989 & 0.392 \\ -0.002 & 0.914 \end{bmatrix}$$

is close to the true minimal transition matrix

$$\mathbf{A}_{\min} := \begin{bmatrix} 0.989 & 0.392 \\ 0.000 & 0.950 \end{bmatrix}$$

with relative error $\|\mathbf{G}_{\min}^+ \hat{\mathbf{B}} \mathbf{G}_{\min} - \mathbf{A}_{\min}\|_F / \|\mathbf{A}_{\min}\|_F = 0.0254$, which is lower than the relative error 0.937 for the full state. Table 1 summarizes these findings in this particular application.

E Parameter Values for the 100-Household Example

Table 2 lists parameter values used in the 100-household Gorman economy of Section 3.3. Notation follows the main text. In particular, Λ_s , Π_s , Δ_h , and Θ_h appear in the service equations following (19), while Δ_k , Θ_k , and Γ appear in the planner's constraints following (21).

Table 2: Parameter values for the 100-household example

<i>Preferences and technology (scalar)</i>		
β	Discount factor	0.952
Λ_s	Service-habit loading	0
Π_s	Service-consumption loading	1
Δ_h	Durable persistence	0
Θ_h	Durable-consumption loading	0
Δ_k	Capital persistence	0.95
Θ_k	Capital-investment loading	1
Γ	Capital productivity	0.1
<i>Aggregate endowment process</i>		
ρ_1	AR(1) coefficient	0.95
σ_a	Aggregate shock std. dev.	0.5
<i>Household heterogeneity</i>		
J	Number of households	100
α_j	Mean endowment	$\sim \mathcal{U}[3, 5]$
ϕ_j	Aggregate exposure	$\sim \mathcal{U}[0.5, 1.5], \sum_j \phi_j = 1$
<i>Idiosyncratic endowment shocks</i>		
J_a	Absorbing households	50
σ_j	Idio. shock std. dev.	$0.2 + 4.8 p_j^2$
ρ_j^d	Idio. shock persistence	$0.9 p_j$

Notes: $p_j := 1 - \text{percentile rank of } \alpha_j$. Idiosyncratic preference shocks are muted. See Section 3.3 for details.

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